7+1: the octonions and the Clifford algebra of 7+1 spacetime

A real realization of the octonions, and a real 16×16 realization of the Clifford algebra of 7+1 spacetime, built from — and mirroring — the split octonions and the 16×16 Dirac matrices of the notebook 4+4.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1), evaluated Tue 27 Oct 2026 14:15:35
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Abstract

Starting from the real 8×8 matrices τ_A and the real 16×16 Dirac matrices T_A of 4+4.nb (split octonions and the Clifford algebra $Cl(4,4)$), this document and the notebook 7+1.nb construct (i) a real realization of the octonions whose multiplication constants come from exactly the same vielbein formula as the split-octonion constants of 4+4.nb; (ii) real 16×16 generators of the Clifford algebra of 7+1 spacetime (one time, seven space dimensions), built only from the matrices of 4+4.nb; (iii) a proof that in the sign convention of 4+4.nb no real 16×16 realization exists, together with the minimal complexification and a real 32×32 realization; (iv) examples that test everything; (v) in sections 7–10 of the notebook, the Dirac equation of the 16-component quantized fermion field dirac16complex with these real Dirac matrices, its chiral form as two coupled 8-component equations, and the 16-component and chiral Kohn–Sham theories of that field coupled to itself (section 11); (vi) in sections 11–19 of the notebook, gravity on the 7+1 spacetime — curvature, the Lovelock tensors (computed with xAct exactly as in the reference notebook Einstein-Lovelock-7+1-Nash.nb, and independently), generalized frame matrices, the canonical spin connection that couples gravity to dirac16complex, and the primordial, modern and far-future epochs — the nine-channel self-coupling of dirac16complex with its Kohn–Sham theory in gravitational fields, and the Standard Model on the world brane with the portals of dirac16complex and neutrino flavour conversion (section 12). All 310 checks of the notebook are True.

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1. Results at a glance

Octonions from the split-octonion matrices. With $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ (the 8×8 matrix $\mathfrak{o}$ of 4+4.nb),

$$\tau_0^O = 1_8, \qquad \tau_A^O = \tau_A \;\; (A = 1,2,3), \qquad \tau_A^O = \tau_A\,\sigma \;\; (A = 4,5,6,7), \qquad \bar\tau_A^O = (\tau_A^O)^{\mathsf T}.$$

These are real, $\tau_A^O\bar\tau_B^O + \tau_B^O\bar\tau_A^O = 2\,\delta_{AB}\,1_8$, and the vielbein formula of 4+4.nb turns them into the multiplication table of the octonions $\mathbb{O}$ (section 6).

Real Clifford algebra of 7+1 spacetime. With the 16×16 Dirac matrices T_A of 4+4.nb and $S_{44} = T_4T_5T_6T_7$,

$$\Gamma_0^{(1,7)} = T_0, \qquad \Gamma_i^{(1,7)} = T_0\,T_i \;\; (i = 1,2,3), \qquad \Gamma_k^{(1,7)} = T_0\,T_k\,S_{44} \;\; (k = 4,5,6,7),$$

$$\Gamma_A^{(1,7)}\Gamma_B^{(1,7)} + \Gamma_B^{(1,7)}\Gamma_A^{(1,7)} = 2\,\eta_{AB}^{(1,7)}\,1_{16}, \qquad \eta^{(1,7)} = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1).$$

All entries are $0, \pm 1$. Time (x_0, the octonion unit) squares to $+1$, the seven space directions to -1. The quadratic form is the real part of the octonion square, $\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2$; the algebra is $Cl(1,7) \cong \mathbb{R}(16)$ (section 7.2).

The sign convention of 4+4.nb. 4+4.nb writes metrics as "space-like = +1, time-like = −1". In that convention the 7+1 metric is $\eta^{(7,1)} = \mathrm{diag}(-1,+1,\ldots,+1) = -\eta^{(1,7)}$ and the relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}^{(7,1)}1_{16}$ defines $Cl(7,1) \cong \mathbb{H}(8)$, which has NO real 16×16 realization, whatever the order or signs of the indices (theorem of section 7.4). The minimal complexification multiplies only the time block by i and keeps the exact block form of 4+4.nb:

$$\Gamma_A^{(7,1)} = \begin{pmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{pmatrix}, \qquad \tau_0^C = \bar\tau_0^C = i\,1_8, \quad \tau_k^C = \tau_k^O,\; \bar\tau_k^C = \bar\tau_k^O \;\; (k = 1,\ldots,7),$$

$\Gamma^{(7,1)}_A\Gamma^{(7,1)}_B + \Gamma^{(7,1)}_B\Gamma^{(7,1)}_A = 2\,\eta^{(7,1)}_{AB}\,1_{16}$.

A real realization of $Cl(7,1)$ needs 32×32 matrices: $\widehat{\Gamma}^{(7,1)}_A = \varepsilon \otimes \Gamma^{(1,7)}_A$ with $\varepsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (section 7.5).

The Dirac equation of dirac16complex and its Kohn–Sham theories (sections 7–10 of the notebook; section 11 below). With the real $\Gamma^{(1,7)}_A$ and $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma^{(1,7)}_A$, the 16-component field Ψ of complex anticommuting operators obeys $(i\Gamma^A\partial_A - m)\Psi = 0$; its chiral 8-component parts $\Psi_L = (\Psi_\uparrow + i\Psi_\downarrow)/\sqrt{2}$, $\Psi_R = (i\Psi_\uparrow + \Psi_\downarrow)/\sqrt{2}$ obey

$$\big(i\,\partial_0 + \tau^O_i\,\partial_i\big)\Psi_L = m\,\Psi_R, \qquad \big(i\,\partial_0 - \tau^O_i\,\partial_i\big)\Psi_R = m\,\Psi_L,$$

coupled by the mass. The field coupled to itself by scalar, pseudoscalar, vector and axial-vector contact interactions has a 16-component Kohn–Sham theory and a chiral (8+8-component) Kohn–Sham theory with a complex Kohn–Sham mass.

Gravity, the self-coupled field in gravitational fields, and the Standard Model (sections 11–19 of the notebook; section 12 below). On a metric $g_{\mu\nu}(X)$ of signature $(+,-,\ldots,-)$ the Lovelock tensors of order k, with the generalized Kronecker delta $\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det[\delta^{\mu_i}_{\nu_j}]$ and the Riemann tensor $R^{\gamma\delta}{}_{\alpha\beta}$, are

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\,R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k} \qquad (E^{(1)} = G,\ E^{(4)} \equiv 0 \text{ in eight dimensions}),$$

and the generalized frame matrices $\Gamma_\mu(X) = e^A{}_\mu(X)\Gamma^{(1,7)}_A$ with the canonical spin connection Ω_μ have a vanishing total covariant derivative, so that dirac16complex couples to gravity through

$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{{}^{\lambda}_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu,\Gamma_\nu] = 0, \qquad \big(i\,\Gamma^\mu(X)(\partial_\mu + \Omega_\mu) - m\big)\Psi = 0.$$

The field coupled to itself through all nine ranks of the Clifford algebra has a Kohn–Sham theory in gravitational fields, solved in the three cosmological epochs with its back-reaction on the expansion; the Standard Model lives on the brane $x_4 = \cdots = x_7 = 0$ and meets dirac16complex through portals. Sections 11–19 contain 144 checks of their own.

Verification. The notebook 7+1.nb contains 310 checks; evaluated in 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026) it returns True for every one of them, with 0 message cells, in 225.3 s. Independent Python programs reproduce every matrix, table and number that the notebook prints (section 9).

2. Coordinates, metrics and conventions

4+4.nb and 7+1.nb use the same eight coordinate symbols $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ (Mathematica: `X = {x0, x1, x2, x3, x4, x5, x6, x7}`; never $x_1, \ldots, x_8$). Their meaning differs:

coordinate	meaning in 4+4.nb (signature 4+4)	meaning in 7+1.nb (signature 7+1)
x_0	hidden space-like dimension	ordinary time; octonion unit $e_0 = 1$
x_1, x_2, x_3	ordinary 3-space	ordinary 3-space; imaginary units e_1, e_2, e_3
x_4	ordinary time	auxiliary space-like; $e_4 = \ell$ (Cayley–Dickson doubling unit)
x_5, x_6, x_7	three auxiliary time-like dimensions ("superluminal deflating time")	auxiliary space-like; e_5, e_6, e_7

Metrics (Mathematica names in brackets):

$\eta^{(4,4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$	[η4488] (4+4.nb: space-like $+1$, time-like -1)
$\eta^{(7,1)} = \mathrm{diag}(-1,+1,+1,+1,+1,+1,+1,+1)$	[η71] (7+1 in the sign convention of 4+4.nb)
$\eta^{(1,7)} = -\eta^{(7,1)} = \mathrm{diag}(+1,-1,\ldots,-1)$	[η17] (7+1, octonionic convention)
$g^O = 1_8$	[g0] (norm of the octonions)

Conventions. Indices $A, B, K, L \in \{0,\ldots,7\}$ label coordinates, octonion units and Clifford generators; $i, j \in \{1,\ldots,7\}$ label imaginary units and space directions; $h \in \{1,2,3\}$. 1_n is the identity matrix, M^{T} the transpose, δ_{AB} the Kronecker delta, $\{X,Y\} = XY + YX$, and $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ a matrix of blocks. The index 8 on a Clifford generator denotes the ordered product of all eight, as in 4+4.nb ($T_8 = T_0T_1\cdots T_7$). $Cl(p,q)$ always means: p generators square to $+1$ and q to -1, i.e. $v^2 = Q(v)$ with Q of signature (p,q). Mathematica lists are 1-based, so the metric entry η_{AB} is written `η[[A+1, B+1]]` in the notebook.

3. Notation used in this document

Every symbol of this document, with its Mathematica name in 7+1.nb and its definition. Names in the first group are reproduced exactly from 4+4.nb (where some of them carried 2-D names made with the Notation package: $\bar\tau_A$ was a symbolized tau-bar, T_A was `T16^A[A]`, E_{44} was `E^A_a[h]`); the plain one-line names avoid confusing them with powers or indices.

symbol	Mathematica	definition / meaning
σ	`σ`	$\left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$, split metric on spinor space (4+4.nb `σ88`)
$s_h,\ t_h$	`s4by4[h]`, `t4by4[h]`	self-dual / anti-self-dual 4×4 matrices, $(s_h)_{pq} = \epsilon_{hpq4} - (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$, $(t_h)_{pq} = \epsilon_{hpq4} + (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$
ζ_A	`ζ[A]`	$\zeta_0 = 1_8$, $\zeta_h = \left(\begin{smallmatrix} 0 & s_h \\ s_h & 0 \end{smallmatrix}\right)$, $\zeta_{7-h} = \left(\begin{smallmatrix} 0 & t_h \\ -t_h & 0 \end{smallmatrix}\right)$, $\zeta_7 = \zeta_1 \cdots \zeta_6$
$\tau_A,\ \bar\tau_A$	`τ[A]`, `τbar[A]`	$\tau_A = \zeta_A$, $\bar\tau_A = \sigma\tau_A^{\mathsf T}\sigma$ (split octonions)
T_A	`T16[A]`	$\left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$, Dirac matrices of 4+4.nb; $T_8 = T_0 \cdots T_7$
$P_L,\ P_R$	`PL44`, `PR44`	$(1_{16} \mp T_8)/2$
u	`unit`	$(1,0,0,0,1,0,0,0)/\sqrt{2}$, the unit spinor (`uBasis[[8]]`, `hUSE = 8`)
E_{44}	`E44[8]`	split vielbein, rows $u^{\mathsf T}\sigma\bar\tau_B$
ε_B	`eA[B]`	split-octonion units, $B = 1, \ldots, 8$
$\tau^O_A,\ \bar\tau^O_A$	`τO[A]`, `τObar[A]`	octonion matrices, section 6.1
E^O	`EO`	octonion vielbein, rows $u^{\mathsf T}\bar\tau^O_B$
L^O_A	`LO[A]`	$E^O\tau^O_A(E^O)^{\mathsf T}$, left multiplication by e_A
e_A	`e[A]`	octonion units, $A = 0, \ldots, 7$, $e_0 = 1$
$N(x),\ \bar x,\ \mathrm{Re}\,x$	`oNorm2`, `oConj`, `oRe`	norm, conjugate, real part
T^O_A	`T16O[A]`	$\left(\begin{smallmatrix} 0 & \bar\tau^O_A \\ \tau^O_A & 0 \end{smallmatrix}\right)$, generators of $Cl(8,0)$
S_{44}	`S44`	$T_4T_5T_6T_7 = \mathrm{diag}(\sigma,\sigma)$
$\Gamma^{(1,7)}_A$	`Γ17[A]`	real generators of the 7+1 Clifford algebra, metric $\eta^{(1,7)}$
$\Gamma^{(7,1)}_A$	`Γ71[A]`	complex generators, metric $\eta^{(7,1)}$
$\widehat\Gamma^{(7,1)}_A$	`Γ71R32[A]`	real 32×32 generators, metric $\eta^{(7,1)}$
$\Gamma^{(5,3)}_A,\ \Gamma^{(0,8)}_A$	`Γ53[A]`, `Γ08[A]`	real generators of $Cl(5,3)$ and $Cl(0,8)$ (section 8.6)
Σ_{AB}	`Σ17[A,B]`	$(\Gamma^{(1,7)}_A\Gamma^{(1,7)}_B - \Gamma^{(1,7)}_B\Gamma^{(1,7)}_A)/4$
ε	`ε2`	$\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$
$\sigma_1,\ \sigma_3$	—	$\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$
$\Gamma^{(1,7)}_8$	`Γ17[8]`	$\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1\cdots\Gamma^{(1,7)}_7 = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, the volume element
$Q_i,\ Q_j,\ Q_k$	`Qi`, `Qj`, `Qk`	$\varepsilon \otimes 1_{16}$, $\sigma_1 \otimes \Gamma^{(1,7)}_8$, $\sigma_3 \otimes \Gamma^{(1,7)}_8$ (quaternion units, section 7.5)
$\chi^{(7,1)},\ \chi^{(1,7)}$	`χ71`, `χ17`	$-i\,\Gamma^{(7,1)}_8$ and $i\,\Gamma^{(1,7)}_8$, chirality matrices (section 7.6)
$\vec x$	`xImag`	imaginary part $x_1e_1 + \cdots + x_7e_7$ of the octonion x (section 7.7)
$S(\beta),\ R(\theta),\ \Lambda$	`Sβ`, `Rθ`, `Λboost`, `Λrot`	a boost, a rotation and their Lorentz matrices (section 8.2)
Γ^A	`Γ17up[A]`	$\eta^{AA}_{(1,7)}\Gamma^{(1,7)}_A$, index raised with the inverse metric
$p,\ m,\ \not p$	`pvec`, `mass`, `pslash`	$p = (3,2,2,0,0,0,0,0)$, $m = 1$, $\not p = \sum_A p^A\Gamma^{(1,7)}_A$ (section 8.4)
W	`realAmplitudes`	the 8 real solutions w of $(\not p - m)w = 0$, one per row
$k,\ \not k$	`kvec`, `kslash`	$k = (1,1,0,0,0,0,0,0)$, $\not k = \sum_A k^A\Gamma^{(1,7)}_A$ (section 8.5)
$\widehat\Gamma^A\partial_A,\ P$	`DiracOp71R32`, `pslash32`	$\sum_A \eta^{AA}_{(7,1)}\widehat\Gamma^{(7,1)}_A\partial_A$ and $P = \sum_A p^A\widehat\Gamma^{(7,1)}_A$ (section 8.5)
$\Psi = (\Psi_1, \ldots, \Psi_{16})^{\mathsf T}$	`dirac16complex`	section 11: the quantized 16-component field, complex anticommuting (Grassmann-odd) operators; also, in section 8.3, 16 arbitrary functions
$\bar\Psi,\ \Psi^c$	—	$\Psi^\dagger\Gamma^0$ (Dirac adjoint); $\Gamma^{(1,7)}_8\Psi^*$ (charge conjugate)
$\Psi_\uparrow,\ \Psi_\downarrow$	—	upper and lower 8 components of Ψ
$\Psi_L,\ \Psi_R$	`ϕL`, `ϕR`	$(\Psi_\uparrow + i\Psi_\downarrow)/\sqrt{2}$, $(i\Psi_\uparrow + \Psi_\downarrow)/\sqrt{2}$, chiral 8-component fields
$P^{(1,7)}_L,\ P^{(1,7)}_R$	`PL17`, `PR17`	$\frac{1}{2}(1_{16} \mp \chi^{(1,7)})$, the complex chiral projections of 7+1 spacetime
χ	`χ17`	in section 11: $\chi = \chi^{(1,7)} = i\Gamma^{(1,7)}_8$
m	`mD`, `mass`	mass ($m = 1$ in the examples)
$\vec p,\ E$	`pv7`, `Ep`	momentum $(p^1, \ldots, p^7)$; $E = \sqrt{m^2 + \vec p^{\,2}}$
$H_{\mathrm D}$	`hD[ψ, m]`	$-i\Gamma^0\Gamma^i\partial_i + m\Gamma^0$, the Dirac Hamiltonian: $i\partial_0\Psi = H_{\mathrm D}\Psi$
$e_s,\ w_s$	`UnitVector[8, s]`, `uS[s, M, E]`	unit column of length 8; $w_s = \frac{1}{\sqrt{E}}(E\,e_s;\ (m1_8 - p^i\tau^O_i)e_s)$, real positive-energy amplitude, $s = 1, \ldots, 8$

symbol	Mathematica	definition / meaning
g_S, g_P, g_V, g_A	`gS`, `gP`, `gV`, `gA`	real couplings of the scalar, pseudoscalar, vector and axial-vector contact interactions (mass dimension -6)
s, p, v_A, a_A	—	real external sources
ρ_S, ρ_P, j^A, j_5^A	`dens16[M]`	$\bar\Psi\Psi$, $\bar\Psi i\chi\Psi$, $\bar\Psi\Gamma^A\Psi$, $\bar\Psi\Gamma^A\chi\Psi$, or the sums over Kohn–Sham orbitals; $j_A = \eta^{(1,7)}_{AA} j^A$
Φ_k, λ_k	`orbV`	Kohn–Sham orbitals (16 components) and their eigenvalues
Φ_{kL}, Φ_{kR}	`orbLV`, `orbRV`	chiral components of Φ_k, formed like Ψ_L, Ψ_R
H_{KS}	`hKS[p]`	Kohn–Sham Hamiltonian (section 11.3)
$\mathcal{S}$, $\mathcal{P}$, $\mathcal{V}_A$, $\mathcal{A}_A$	`Sks`, `Pks`, `VKS`, `AKS`	Kohn–Sham potentials (lower index A)
F, T_s, E_H, E_{xc}, E_x	`EH16`, `Exfun`	universal functional; kinetic energy of the orbitals; Hartree, exchange-correlation and exchange energies
$\mathcal{R}$	`Rorb`	$\sum_k \Phi_k\bar\Phi_k$, 16×16
κ	`κc`	$\sum_k \Phi^\dagger_{kL}\Phi_{kR}$, complex scalar density
$\mathcal{M}$, $\mathcal{V}^L_A$, $\mathcal{V}^R_A$	`Mks`, `VLks`, `VRks`	$m+\mathcal{S}+i\mathcal{P}$ (complex Kohn–Sham mass); $\mathcal{V}_A-\mathcal{A}_A$; $\mathcal{V}_A+\mathcal{A}_A$
ϑ	`Uθ`	angle of a chiral rotation $e^{i\vartheta\chi}$
k_F, m^*, ρ, E/A, E_F	`kF0`, `gap7`, `rho7`, `eps7`, `fermi7`	uniform matter: Fermi momentum, effective mass, density j^0, energy per particle, Fermi energy
Λ_{c}, g, g_c	`Λc`, `gNJL`, `gcrit`	momentum cutoff of the Dirac sea; $g=g_S=g_P$; critical coupling $-90\pi^4/\Lambda_{\mathrm{c}}^6$
$g_{\mu\nu}$, $\{^\lambda_{\mu\nu}\}$	`cv["g"]`, `cv["chr"]`	section 12: metric (signature $(+,-,\dots,-)$) and Christoffel symbols (never written Γ)
$R^\rho{}_{\sigma\mu\nu}$, $R_{\sigma\nu}$, R, $G^\mu{}_\nu$	`cv["riem"]`, `cv["ric"]`, `cv["R"]`, `cv["G"]`	Riemann (Landau–Lifshitz), Ricci, scalar curvature, Einstein tensor
$E^{(k)\mu}{}_\nu$, L_k	`Lovelock1/2/3`, `lovelockE`, `lovelockL`	Lovelock tensors and densities ($k=1,2,3$)
α_k, $\bar\Lambda$, κ	`alf[k]`, `lamb`, `kap`	Lovelock couplings ($\alpha_1=1$), cosmological constant, $8\pi G_8$
w_k, Λ, H	`w1`, `w2`, `w3`, `Λ`, `H`	pure-number couplings and inverse length of the reference notebook
a, b, H_a, H_b	`a3[x0]`, `b4[x0]`, `Ha`, `Hb`	scale factors of ordinary and auxiliary space; Hubble rates a'/a, b'/b
$e^A{}_\mu$, $\Gamma_\mu(X)$, Ω_μ	`fd["e"]`, `fd["Gam"]`, `fd["Om"]`	frame field, generalized frame matrices, spinor connection
Θ^K, I_r, g_r	`0up[K]`, `Ichan`, `gg[r]`	Clifford products of rank r, their invariants and couplings
F_{rs}, G_s	`fierzF`, `Geff`	Fierz matrix and Hartree–Fock couplings $G=(1-F^{\mathsf{T}})g$
M, ε, p_a, p_b	`ksStress`	effective mass, energy density and pressures of Kohn–Sham matter
W (brane), $h_{\mu\nu}$	—	world brane $x_4=\cdots=x_7=0$, induced metric
v, V_{CKM}, U_{PMNS}	`vEW`, `vCKM`, `uPMNS`	Higgs vacuum value, quark and lepton mixing matrices
R_A, U	`rBr`, `uInt`	brane form of the Dirac matrices and the intertwiner $\Gamma^{(1,7)}_A U = U R_A$
$P(\alpha\to\beta)$	`pVac`, `pMat`	neutrino flavour-conversion probability

4. The two definitions, and four corrections

4.1 [1A] The octonions

The octonions $\mathbb{O}$ form an 8-dimensional, non-associative, non-commutative normed division algebra over $\mathbb{R}$, with basis $e_0=1, e_1,\dots,e_7$:

$$x=\textstyle\sum_{A=0}^7 x_A e_A, \qquad e_i^2=-1\ (i=1,\dots,7), \qquad e_ie_j=-e_je_i\ (i\neq j),$$
$$\bar x = x_0-\textstyle\sum_{i=1}^7 x_ie_i, \qquad \overline{xy}=\bar y\,\bar x, \qquad \mathrm{Re}(x)=\tfrac12(x+\bar x)=x_0,$$
$$N(x)=x\bar x=\bar x x=x_0^2+x_1^2+\cdots+x_7^2, \qquad N(xy)=N(x)N(y), \qquad x^{-1}=\bar x/N(x),$$
$$x(xy)=(xx)y, \qquad (yx)x=y(xx) \qquad \text{(alternativity; power-associativity follows).}$$

Products of distinct imaginary units follow the oriented lines of the Fano plane, each line spanning a quaternion subalgebra. Cayley–Dickson: $\mathbb{O}=\mathbb{H}\oplus\mathbb{H}$ as pairs of quaternions.

4.2 [1B] The Clifford algebra of an 8-dimensional space

$$
\begin{aligned}
Cl(V,Q) = \mathcal{T}(V)/\mathcal{I}_Q(V), \qquad & \mathcal{I}_Q(V) = \big\langle v\otimes v - Q(v)1 \big\rangle, \\
v^2 = Q(v)\,1, \qquad & vw + wv = 2B(v,w)\,1, \qquad B(v,w) = \tfrac{1}{2}\big(Q(v+w) - Q(v) - Q(w)\big), \\
e_A^2 = Q(e_A), \qquad & e_A e_B = -e_B e_A \ (A \neq B), \qquad \dim Cl(V,Q) = 2^8 = \textstyle\sum_{k=0}^{8} \binom{8}{k} = 256, \\
Cl_8(\mathbb{C}) \cong \mathbb{C}(16). &
\end{aligned}
$$

4.3 Corrections (each verified in 7+1.nb)

1. **Cayley–Dickson formula.** The second formula of [1A], $(a,b)(c,d) = (ac - \bar{d}b,\ \bar{a}d + cb)$, does not define the octonions: it is not alternative and its norm is not multiplicative. Counterexample: $x = (1,1) = e_0 + e_4$, $y = (i,0) = e_1$ give $x^2 = (0,2)$, $x^2 y = (0,2i)$, but $xy = (i,i)$ and $x(xy) = (2i,2i)$. A correct formula of the same shape is $(a,b)(c,d) = (ac - d\bar{b},\ \bar{a}d + cb)$ (J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002), section 2.2); the octonions of this document satisfy it exactly with $x = a + \ell b$, $\ell = e_4$. The first formula of [1A] contains a stray symbol "x" and cannot be read unambiguously.
2. **Grade 4.** $\binom{8}{4} = 70$, not 72: $1 + 8 + 28 + 56 + 70 + 56 + 28 + 8 + 1 = 256$.
3. **Classification.** With $v^2 = Q(v)$ and p = number of generators squaring to $+1$, the correct table is the one below. $\mathbb{C}(16)$ can never occur: its real dimension is $2 \cdot 16^2 = 512$, while every $Cl(p,q)$ with $p + q = 8$ has real dimension $2^8 = 256$. Several "Metric Signs" entries of [1B] were also mistyped.
4. **Split octonions are not a division algebra.** In the split-octonion table of 4+4.nb, $(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = \varepsilon_1 - \varepsilon_5^2 = 0$ although both factors are nonzero. They form a composition algebra with an indefinite norm of signature (4,4).

$Cl(p,q)$	**squares of the generators**	**correct**	**claimed in [1B]**	**status**
$Cl(8,0)$	+,+,+,+,+,+,+,+	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees
$Cl(7,1)$	+,+,+,+,+,+,+,−	$\mathbb{H}(8)$	$\mathbb{R}(16)$	wrong in [1B]
$Cl(6,2)$	+,+,+,+,+,+,−,−	$\mathbb{H}(8)$	$\mathbb{H}(8)$	agrees (signs mistyped)
$Cl(5,3)$	+,+,+,+,+,−,−,−	$\mathbb{R}(16)$	$\mathbb{C}(16)$	wrong in [1B]
$Cl(4,4)$	+,+,+,+,−,−,−,−	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees (signs mistyped)
$Cl(3,5)$	+,+,+,−,−,−,−,−	$\mathbb{H}(8)$	$\mathbb{C}(16)$	wrong in [1B]
$Cl(2,6)$	+,+,−,−,−,−,−,−	$\mathbb{H}(8)$	$\mathbb{H}(8)$	agrees (signs mistyped)
$Cl(1,7)$	+,−,−,−,−,−,−,−	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees
$Cl(0,8)$	−,−,−,−,−,−,−,−	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees

Consequence for this project. With $v^2 = Q(v)$ and the sign convention of 4+4.nb ($Q = \eta^{(7,1)}$) the Clifford algebra of 7+1 spacetime is $Cl(7,1) \cong \mathbb{H}(8)$: no real 16×16 matrices exist (section 7.4). With the opposite overall sign ($Q = \eta^{(1,7)}$, the octonions' own form $\mathrm{Re}(xx)$) it is $Cl(1,7) \cong \mathbb{R}(16)$, realized by the real matrices $\Gamma_A^{(1,7)}$ built from 4+4.nb (section 7.2). Both describe the same spacetime: one time and seven space dimensions.

5. Recap: the 4+4 construction, reproduced exactly from 4+4.nb

5.1 The 8×8 matrices σ, ζ_A, τ_A, $\bar{\tau}_A$

With ϵ_{hpq4} the sign of the permutation $(h,p,q,4)$ (Mathematica `Signature[{h, p, q, 4}]`), 4+4.nb defines the self-dual and anti-self-dual antisymmetric 4×4 matrices

$$
(s_h)_{pq} = \epsilon_{hpq4} - \big(\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}\big), \qquad (t_h)_{pq} = \epsilon_{hpq4} + \big(\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}\big), \qquad h = 1,2,3,\ p,q = 1,\ldots,4,
$$

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix},\ s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},\ s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},\ t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},\ t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

and from them the 8×8 matrices

$$\sigma = \begin{pmatrix} 0 & 1_4 \\ 1_4 & 0 \end{pmatrix}, \qquad \zeta_0 = 1_8, \qquad \zeta_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \qquad \zeta_{7-h} = \begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}\ (h=1,2,3),$$

$$\zeta_7 = \zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6 = \mathrm{diag}(-1_4, 1_4), \qquad \tau_A = \zeta_A, \qquad \bar\tau_0 = 1_8, \quad \bar\tau_A = \sigma\,\tau_A^{\mathsf T}\sigma\ (A=1,\dots,7).$$

$$\tau_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_7 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

They satisfy (all verified in 7+1.nb exactly as in 4+4.nb)

$$\tfrac{1}{2}\big(\zeta_A\zeta_B + \zeta_B\zeta_A\big) = -\eta^{(4,4)}_{AB}\,1_8 \quad (A,B=1,\dots,7), \qquad -\eta^{(4,4)}_{AA}\,\zeta_A = \zeta_A^{\mathsf T},$$
$$\tfrac{1}{2}\big(\tau_A\bar\tau_B + \tau_B\bar\tau_A\big) = \tfrac{1}{2}\big(\bar\tau_A\tau_B + \bar\tau_B\tau_A\big) = \eta^{(4,4)}_{AB}\,1_8 \quad (A,B=0,\dots,7),$$

$$\tau_1\tau_2\cdots\tau_7 = 1_8, \qquad \sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7, \qquad \sigma^2 = 1_8,$$
$$\sigma\tau_h = \tau_h\sigma \ (h=1,2,3), \qquad \sigma\tau_k = -\tau_k\sigma \ (k=4,\ldots,7).$$

So τ_1,τ_2,τ_3 are antisymmetric and square to -1_8; $\tau_4,\ldots,\tau_7$ are symmetric and square to $+1_8$: they are the left multiplications of the three imaginary and the four split units of the split octonions.

5.2 The 16×16 Dirac matrices T_A of 4+4.nb

$$T_A = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix} \ (A=0,\ldots,7), \qquad T_AT_B + T_BT_A = 2\,\eta^{(4,4)}_{AB}\,1_{16},$$
$$T_8 = T_0T_1\cdots T_7 = \mathrm{diag}(-1_8, 1_8), \qquad \sigma_{16} = T_0T_1T_2T_3 = \mathrm{diag}(-\sigma,\sigma),$$
$$P_L = \tfrac{1}{2}(1_{16} - T_8) = \mathrm{diag}(1_8, 0), \qquad P_R = \tfrac{1}{2}(1_{16} + T_8) = \mathrm{diag}(0, 1_8).$$

5.3 How the split-octonion multiplication constants come from τ_A

Choose the unit spinor $u = (1,0,0,0,1,0,0,0)^{\mathsf T}/\sqrt{2}$, an eigenvector of σ: $\sigma u = u$, $u^{\mathsf T}u = u^{\mathsf T}\sigma u = 1$. The vielbein E_{44} (4+4.nb: `E^A_a[hUSE]`) and its inverse are

$$(E_{44})_{Ba} = \left(u^{\mathsf T}\sigma\,\bar\tau_B\right)_a, \qquad E^{-1}_{44} = (u^{\mathsf T}\sigma u)\left(\eta^{(4,4)}_{00}\tau_0 u,\ \eta^{(4,4)}_{11}\tau_1 u,\ \ldots,\ \eta^{(4,4)}_{77}\tau_7 u\right) \quad \text{(columns)}.$$

The left multiplication by the split unit ε_B (4+4.nb: `eA[B]`, $B=1,\ldots,8$) is the matrix $g_{BB}\,E_{44}\,\tau_{B-1}\,E^{-1}_{44}$ with $g = \eta^{(4,4)}$:

$$\varepsilon_B\,\varepsilon_C = \sum_{D=1}^{8}\left(g_{BB}\,E_{44}\,\tau_{B-1}\,E^{-1}_{44}\right)_{DC}\varepsilon_D.$$

In words: the split-octonion multiplication constants are the matrix elements of the 8×8 matrices τ_A, written in the octonion basis by the vielbein, times the metric sign g_{BB}. This is the table `gridx` of 4+4.nb (row ε_B times column ε_C):

·	ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7	ε_8
ε_1	ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7	ε_8
ε_2	ε_2	$-\varepsilon_1$	$-\varepsilon_4$	ε_3	$-\varepsilon_6$	ε_5	ε_8	$-\varepsilon_7$
ε_3	ε_3	ε_4	$-\varepsilon_1$	$-\varepsilon_2$	$-\varepsilon_7$	$-\varepsilon_8$	ε_5	ε_6
ε_4	ε_4	$-\varepsilon_3$	ε_2	$-\varepsilon_1$	$-\varepsilon_8$	ε_7	$-\varepsilon_6$	ε_5
ε_5	ε_5	ε_6	ε_7	ε_8	ε_1	ε_2	ε_3	ε_4
ε_6	ε_6	$-\varepsilon_5$	ε_8	$-\varepsilon_7$	$-\varepsilon_2$	ε_1	$-\varepsilon_4$	ε_3
ε_7	ε_7	$-\varepsilon_8$	$-\varepsilon_5$	ε_6	$-\varepsilon_3$	ε_4	ε_1	$-\varepsilon_2$
ε_8	ε_8	ε_7	$-\varepsilon_6$	$-\varepsilon_5$	$-\varepsilon_4$	$-\varepsilon_3$	ε_2	ε_1

Split octonions: $\varepsilon_1 = 1$, $\varepsilon_2^2 = \varepsilon_3^2 = \varepsilon_4^2 = -1$, $\varepsilon_5^2 = \cdots = \varepsilon_8^2 = +1$, norm $N_S(x) = x^{\mathsf T}\eta^{(4,4)}x$ with $N_S(xy) = N_S(x)N_S(y)$, alternative, and with zero divisors: $(\varepsilon_1+\varepsilon_5)(\varepsilon_1-\varepsilon_5) = 0$.

6. The octonions: a real realization mirroring the split octonions

6.1 From τ to τ^O

All seven imaginary octonion units must square to -1, but $\tau_4,\ldots,\tau_7$ square to $+1$. Because σ commutes with τ_1,τ_2,τ_3, anticommutes with $\tau_4,\ldots,\tau_7$ and squares to 1_8,

$$(\tau_k\sigma)(\tau_k\sigma) = -\tau_k\tau_k\,\sigma\sigma = -1_8 \qquad (k=4,\ldots,7),$$

and all anticommutation relations survive. Over $\mathbb{C}$ one would multiply τ_k by i; the real matrix σ replaces that i. Definition:

$$\tau^O_0 = 1_8, \qquad \tau^O_A = \tau_A \ (A=1,2,3), \qquad \tau^O_A = \tau_A\,\sigma \ (A=4,\ldots,7), \qquad \bar\tau^O_A = g^O(\tau^O_A)^{\mathsf T}g^O = (\tau^O_A)^{\mathsf T}.$$

$$\tau_4^O=\begin{pmatrix}0&1&0&0&0&0&0&0\\-1&0&0&0&0&0&0&0\\0&0&0&-1&0&0&0&0\\0&0&1&0&0&0&0&0\\0&0&0&0&0&-1&0&0\\0&0&0&0&1&0&0&0\\0&0&0&0&0&0&0&1\\0&0&0&0&0&0&-1&0\end{pmatrix},\quad \tau_5^O=\begin{pmatrix}0&0&-1&0&0&0&0&0\\0&0&0&-1&0&0&0&0\\1&0&0&0&0&0&0&0\\0&1&0&0&0&0&0&0\\0&0&0&0&0&0&1&0\\0&0&0&0&0&0&0&1\\0&0&0&0&-1&0&0&0\\0&0&0&0&0&-1&0&0\end{pmatrix},$$

$$\tau_6^O=\begin{pmatrix}0&0&0&-1&0&0&0&0\\0&0&1&0&0&0&0&0\\0&-1&0&0&0&0&0&0\\1&0&0&0&0&0&0&0\\0&0&0&0&0&0&0&1\\0&0&0&0&0&0&-1&0\\0&0&0&0&0&1&0&0\\0&0&0&0&-1&0&0&0\end{pmatrix},\quad \tau_7^O=\begin{pmatrix}0&0&0&0&-1&0&0&0\\0&0&0&0&0&-1&0&0\\0&0&0&0&0&0&-1&0\\0&0&0&0&0&0&0&-1\\1&0&0&0&0&0&0&0\\0&1&0&0&0&0&0&0\\0&0&1&0&0&0&0&0\\0&0&0&1&0&0&0&0\end{pmatrix},$$

and $\tau_1^O=\tau_1$, $\tau_2^O=\tau_2$, $\tau_3^O=\tau_3$ as displayed in section 5.1. Properties (verified):

$$\begin{aligned}(\tau_i^O)^{\mathsf T}=-\tau_i^O,\qquad (\tau_i^O)^2=-1_8,\qquad \tau_i^O\tau_j^O=-\tau_j^O\tau_i^O\ (i\neq j),\\ \tau_A^O\bar\tau_B^O+\tau_B^O\bar\tau_A^O=\bar\tau_A^O\tau_B^O+\bar\tau_B^O\tau_A^O=2\,g_{AB}^O\,1_8=2\,\delta_{AB}\,1_8,\\ \tau_1^O\tau_2^O\cdots\tau_7^O=1_8\qquad(\text{as }\tau_1\cdots\tau_7=1_8\text{ in 4+4.nb}),\\ \tau_4^O=\tau_5\tau_6\tau_7,\quad \tau_5^O=-\tau_4\tau_6\tau_7,\quad \tau_6^O=\tau_4\tau_5\tau_7,\quad \tau_7^O=-\tau_4\tau_5\tau_6.\end{aligned}$$

For $x=\sum_A x_A e_A$ put $\tau^O(x)=\sum_A x_A\tau_A^O$; then $\tau^O(x)\,\bar\tau^O(x)=N(x)\,1_8$ with $N(x)=x_0^2+\cdots+x_7^2$, the matrix form of $x\bar x=N(x)$. In 4+4.nb the same relation holds with $\eta^{(4,4)}$, the split norm.

6.2 The octonion vielbein (same unit spinor u)

In 4+4.nb the rows of the vielbein are $u^{\mathsf T}\sigma\bar\tau_B$, with the split spinor metric σ. For the octonions the spinor metric is Euclidean ($\sigma\to g^O=1_8$), so

$$(E^O)_{Ba}=\big(u^{\mathsf T}g^O\bar\tau_B^O\big)_a,\qquad (E^O)^{-1}=(E^O)^{\mathsf T}=(u^{\mathsf T}g^Ou)\big(g_{00}^O\tau_0^Ou,\dots,g_{77}^O\tau_7^Ou\big),\qquad E^O=\eta^{(4,4)}E_{44}.$$

Because $\sigma u=u$, $\tau_A^Ou=\tau_Au$ for every A: the two vielbeins differ only by the signs of rows 5–8.

$$E^O=\begin{pmatrix}\frac{\sqrt2}{2}&0&0&0&\frac{\sqrt2}{2}&0&0&0\\0&0&0&-\frac{\sqrt2}{2}&0&0&0&-\frac{\sqrt2}{2}\\0&0&\frac{\sqrt2}{2}&0&0&0&\frac{\sqrt2}{2}&0\\0&-\frac{\sqrt2}{2}&0&0&0&-\frac{\sqrt2}{2}&0&0\\0&-\frac{\sqrt2}{2}&0&0&0&\frac{\sqrt2}{2}&0&0\\0&0&\frac{\sqrt2}{2}&0&0&0&-\frac{\sqrt2}{2}&0\\0&0&0&\frac{\sqrt2}{2}&0&0&0&-\frac{\sqrt2}{2}\\-\frac{\sqrt2}{2}&0&0&0&\frac{\sqrt2}{2}&0&0&0\end{pmatrix}.$$

6.3 The octonion multiplication constants

Exactly the formula of 4+4.nb, with $\tau\to\tau^O$, $E_{44}\to E^O$, $g\to g^O$ and a 0-based basis $e_A\leftrightarrow\tau_A^O\leftrightarrow x_A$:

$$L_A^O=g_{AA}^O\,E^O\,\tau_A^O\,(E^O)^{-1},\qquad e_A\,e_K=\sum_{L=0}^{7}\big(L_A^O\big)_{LK}\,e_L,\qquad (xy)_L=\sum_{A,K}x_A\,y_K\,\big(L_A^O\big)_{LK}.$$

Every L_A^O is a signed permutation matrix, so every multiplication constant is 0 or ±1. The table (row e_A times column e_K; Mathematica `gridO`, the mirror of `gridx`):

$\cdot$	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	$-e_3$	e_2	$-e_5$	e_4	e_7	$-e_6$
e_2	e_2	e_3	$-e_0$	$-e_1$	$-e_6$	$-e_7$	e_4	e_5
e_3	e_3	$-e_2$	e_1	$-e_0$	$-e_7$	e_6	$-e_5$	e_4
e_4	e_4	e_5	e_6	e_7	$-e_0$	$-e_1$	$-e_2$	$-e_3$
e_5	e_5	$-e_4$	e_7	$-e_6$	e_1	$-e_0$	e_3	$-e_2$
e_6	e_6	$-e_7$	$-e_4$	e_5	e_2	$-e_3$	$-e_0$	e_1
e_7	e_7	e_6	$-e_5$	$-e_4$	e_3	e_2	$-e_1$	$-e_0$

6.4 The properties of [1A], verified

Units. e_0 is a two-sided identity, $e_i^2 = -e_0$ and $e_i e_j = -e_j e_i$ for $i \neq j$ $(i, j = 1, \ldots, 7)$.

Fano plane. Each product of two distinct imaginary units is $\pm$ a third one. The seven triples are the lines of the Fano plane; written in the cyclic order (a, b, c) with $e_a e_b = +e_c$ (and then $e_b e_c = e_a$, $e_c e_a = e_b$):

$$(1,3,2),\ (1,5,4),\ (1,6,7),\ (2,6,4),\ (2,7,5),\ (3,5,6),\ (3,7,4).$$

Every pair of distinct imaginary units lies on exactly one line, and each line together with e_0 spans a copy of the quaternions (closed and associative). In a picture: put e_1 at the centre, e_2, e_4, e_7 at the vertices of a triangle and e_6, e_3, e_5 at the midpoints of the sides e_2e_4, e_4e_7, e_7e_2; the lines are the three sides, the three medians and the inscribed circle through e_3, e_5, e_6.

Algebraic laws (proved symbolically for general octonions x, y, z with 8 symbolic components each; the Moufang identities, being linear in x and y, are proved for basis units x, y and symbolic z):

$$\begin{aligned}
x(xy) = (xx)y, \qquad (yx)x = y(xx), \qquad (xy)x = x(yx), \qquad x(xx) = (xx)x,\\
[x,y,z] := (xy)z - x(yz) = -[y,x,z] = -[x,z,y],\\
z\big(x(zy)\big) = \big((zx)z\big)y, \qquad x\big(z(yz)\big) = \big((xz)y\big)z, \qquad (zx)(yz) = \big(z(xy)\big)z,\\
x\bar{x} = \bar{x}x = N(x)\,e_0, \qquad \overline{xy} = \bar{y}\,\bar{x}, \qquad \mathrm{Re}(x) = \tfrac{1}{2}(x + \bar{x}) = x_0,\\
N(xy) = N(x)\,N(y), \qquad x\,x^{-1} = x^{-1}x = e_0 \ \text{ for } \ x^{-1} = \bar{x}/N(x),\\
x^2 - 2\,\mathrm{Re}(x)\,x + N(x)\,e_0 = 0.
\end{aligned}$$

The last identity shows that the subalgebra generated by one octonion is $\mathbb{R}$ or $\mathbb{C}$ (power-associativity). Multiplicativity of the norm is Degen's eight-square identity: with $x = \sum a_A e_A$ and $y = \sum b_A e_A$ the components of xy are

$$\begin{aligned}
(xy)_0 &= a_0b_0 - a_1b_1 - a_2b_2 - a_3b_3 - a_4b_4 - a_5b_5 - a_6b_6 - a_7b_7,\\
(xy)_1 &= a_0b_1 + a_1b_0 - a_2b_3 + a_3b_2 - a_4b_5 + a_5b_4 + a_6b_7 - a_7b_6,\\
(xy)_2 &= a_0b_2 + a_1b_3 + a_2b_0 - a_3b_1 - a_4b_6 - a_5b_7 + a_6b_4 + a_7b_5,\\
(xy)_3 &= a_0b_3 - a_1b_2 + a_2b_1 + a_3b_0 - a_4b_7 + a_5b_6 - a_6b_5 + a_7b_4,\\
(xy)_4 &= a_0b_4 + a_1b_5 + a_2b_6 + a_3b_7 + a_4b_0 - a_5b_1 - a_6b_2 - a_7b_3,\\
(xy)_5 &= a_0b_5 - a_1b_4 + a_2b_7 - a_3b_6 + a_4b_1 + a_5b_0 + a_6b_3 - a_7b_2,\\
(xy)_6 &= a_0b_6 - a_1b_7 - a_2b_4 + a_3b_5 + a_4b_2 - a_5b_3 + a_6b_0 + a_7b_1,\\
(xy)_7 &= a_0b_7 + a_1b_6 - a_2b_5 - a_3b_4 + a_4b_3 + a_5b_2 - a_6b_1 + a_7b_0,
\end{aligned}$$

$$\Big(\sum_{A=0}^{7} a_A^2\Big)\Big(\sum_{A=0}^{7} b_A^2\Big) = \sum_{L=0}^{7} (xy)_L^2.$$

Hence $N(x) = 0$ only for $x = 0$, and $xy = 0$ only if $x = 0$ or $y = 0$: no zero divisors, a division algebra (unlike the split octonions).

Non-associativity. Exactly 168 of the 512 ordered basis triples (e_A, e_B, e_K) are non-associative: the triples of distinct imaginary units not on a common Fano line ($7 \cdot 6 \cdot 4 = 168$). Example: $(e_1e_2)e_4 = e_7$ but $e_1(e_2e_4) = -e_7$.

A numerical example. For $x = e_0 + 2e_1 - e_4 + 3e_7$, $y = -2e_0 + e_2 + e_5$, $z = e_3 + e_6$:

$$\begin{aligned} xy &= -2e_0 - 3e_1 + 4e_2 - 2e_3 + 4e_4 - 2e_5 - e_6 - 6e_7, \\ yx &= -2e_0 - 5e_1 - 2e_2 + 2e_3 + 4e_5 + e_6 - 6e_7, \\ N(x) = 15, &\quad N(y) = 6, \quad N(xy) = 90 = N(x)N(y), \\ x^{-1} &= \frac{1}{15}e_0 - \frac{2}{15}e_1 + \frac{1}{15}e_4 - \frac{1}{5}e_7, \\ (xy)z - x(yz) &= -6e_1 - 2e_2 - 6e_3 + 6e_4 + 2e_5 + 6e_6 + 6e_7 \neq 0. \end{aligned}$$

6.5 Cayley–Dickson doubling $\mathbb{O} = \mathbb{H} \oplus \ell\mathbb{H}$

The span of e_0, e_1, e_2, e_3 — the time direction and the three ordinary space directions $x_0, \ldots, x_3$ — is a quaternion subalgebra $\mathbb{H}$, and $\ell = e_4$ doubles it: $e_4 e_k = e_{4+k}$ $(k = 1, 2, 3)$, $e_4^2 = -e_0$. Every octonion is $x = a + \ell b$ with quaternions $a = (x_0, x_1, x_2, x_3)$ and $b = (x_4, x_5, x_6, x_7)$. The orientation inherited from 4+4.nb is $e_1 e_2 = -e_3$ (so $i = e_1$, $j = e_2$, $k = -e_3$ are Hamilton's units). With $\ell^2 = \gamma$ the product is

$$\begin{aligned} (a, b)(c, d) &= \big(ac + \gamma\, d\bar{b},\ \ \bar{a}d + cb\big), \\ \gamma = -1 &: \text{ octonions (this document)}, \qquad \gamma = +1 : \text{ split octonions (4+4.nb)}. \end{aligned}$$

Both cases are proved symbolically in 7+1.nb. The formula $(ac - \bar{d}b,\ \bar{a}d + cb)$ of [1A] fails: for $x = (1, 1)$, $y = (i, 0)$ it gives $x^2 y = (0, 2i)$ but $x(xy) = (2i, 2i)$, and its norm is not multiplicative.

6.6 Octonions versus split octonions

With the split units relabelled 0-based ($\varepsilon_{B+1} \to e_B^S$), the two tables differ only in the products of two doubled units:

$$e_A e_B\big|_{\mathbb{O}} = s_{AB}\; e_A^S e_B^S\big|_{\text{split}}, \qquad s_{AB} = \begin{cases} -1, & A \geq 4 \text{ and } B \geq 4, \\ +1, & \text{otherwise}, \end{cases}$$

because $\ell^2 = -1$ instead of $+1$. Over $\mathbb{C}$ the algebras coincide: $\varphi(e_A^S) = c_A e_A$ with $c = (1, 1, 1, 1, i, i, i, i)$ satisfies $\varphi(xy) = \varphi(x)\varphi(y)$ and $N_{\mathbb{O}}(\varphi(x)) = N_S(x)$, so $\mathbb{O}_{\text{split}} \otimes \mathbb{C} \cong \mathbb{O} \otimes \mathbb{C}$. The real substitute for the factors i is the matrix σ of section 6.1.

7. Clifford algebras from the octonions

7.1 The exact mirror of T_A: the Euclidean algebra $Cl(8, 0)$

Copying $T_A = \left(\begin{smallmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{smallmatrix}\right)$ with $\tau \to \tau^O$:

$$\begin{aligned} &T_A^O = \begin{pmatrix} 0 & \bar{\tau}_A^O \\ \tau_A^O & 0 \end{pmatrix}, \qquad T_A^O T_B^O + T_B^O T_A^O = 2\, g_{AB}^O\, 1_{16} = 2\, \delta_{AB}\, 1_{16} \qquad (Cl(8, 0) \cong \mathbb{R}(16)), \\ &T_A^O = T_A \ (A = 0, 1, 2, 3), \qquad T_k^O = T_k\, S_{44} \ (k = 4, \ldots, 7), \qquad S_{44} = T_4 T_5 T_6 T_7 = \mathrm{diag}(\sigma, \sigma), \\ &T_8^O = T_0^O T_1^O \cdots T_7^O = T_8 = \mathrm{diag}(-1_8, 1_8). \end{aligned}$$

7.2 The real Clifford algebra of 7+1 spacetime: $\Gamma^{(1,7)}$, $Cl(1, 7) \cong \mathbb{R}(16)$

Definition.

$$\Gamma_0^{(1,7)} = T_0^O = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix} \ \text{(time)}, \qquad \Gamma_i^{(1,7)} = T_0^O\, T_i^O = \begin{pmatrix} \tau_i^O & 0 \\ 0 & \bar{\tau}_i^O \end{pmatrix} \ (i = 1, \ldots, 7, \ \text{space}),$$

$$\Gamma_0^{(1,7)} = T_0, \qquad \Gamma_i^{(1,7)} = T_0\, T_i \ (i = 1, 2, 3), \qquad \Gamma_k^{(1,7)} = T_0\, T_k\, S_{44} \ (k = 4, \ldots, 7) \qquad \text{(only 4+4.nb matrices)}.$$

Relations (all verified):

$$\begin{aligned}\Gamma^{(1,7)}_A\Gamma^{(1,7)}_B+\Gamma^{(1,7)}_B\Gamma^{(1,7)}_A &= 2\,\eta^{(1,7)}_{AB}\,1_{16} = -2\,\eta^{(7,1)}_{AB}\,1_{16},\\ (\Gamma^{(1,7)}_0)^{\mathsf T} &= \Gamma^{(1,7)}_0, \quad (\Gamma^{(1,7)}_i)^{\mathsf T} = -\Gamma^{(1,7)}_i,\\ (\Gamma^{(1,7)}_A)^{\mathsf T} &= \Gamma^{(1,7)}_0\,\Gamma^{(1,7)}_A\,\Gamma^{(1,7)}_0 \quad \text{(real Dirac adjoint)},\\ \operatorname{tr}\big(\Gamma^{(1,7)}_{A_1}\cdots\Gamma^{(1,7)}_{A_k}\big) &= 0 \quad (1\le k\le 8,\ A_1<\cdots<A_k).\end{aligned}$$

All entries are 0 or ±1. **Why this is the Clifford algebra of [1B].** (i) For $v=\sum_A v_A e_A$ the matrix $\Gamma^{(1,7)}(v)=\sum_A v_A\Gamma^{(1,7)}_A$ satisfies

$$\Gamma^{(1,7)}(v)^2 = Q(v)\,1_{16}, \qquad Q(v)=v^{\mathsf T}\eta^{(1,7)}v = v_0^2-v_1^2-\cdots-v_7^2,$$

so by the universal property of $Cl(V,Q)=\mathcal{T}(V)/\mathcal{I}_Q(V)$ the map extends to an algebra homomorphism $Cl(V,Q)\to\mathbb{R}(16)$. (ii) The 256 ordered products $\Gamma^{(1,7)}_{A_1}\cdots\Gamma^{(1,7)}_{A_k}$ ($A_1<\cdots<A_k$, $k=0,\dots,8$, counted by $\binom{8}{k}$) are linearly independent, so the homomorphism is injective; both sides have dimension 256, hence $Cl(1,7)\cong\mathbb{R}(16)$. (iii) Only multiples of 1_{16} commute with all $\Gamma^{(1,7)}_A$: the representation is irreducible over $\mathbb{R}$.

7.3 The complexified realization in the exact block form of 4+4.nb: $\Gamma^{(7,1)}$

For the sign convention of 4+4.nb, $\Gamma_A\Gamma_B+\Gamma_B\Gamma_A=2\eta^{(7,1)}_{AB}1_{16}$, complex numbers are unavoidable (section 7.4). The complexification needed is minimal: multiply the time block by i.

$$\tau^C_0=\bar\tau^C_0=i\,1_8, \qquad \tau^C_k=\tau^O_k, \quad \bar\tau^C_k=\bar\tau^O_k\ (k=1,\dots,7), \qquad \Gamma^{(7,1)}_A=\begin{pmatrix}0 & \bar\tau^C_A\\ \tau^C_A & 0\end{pmatrix},$$

$$\Gamma^{(7,1)}_0=i\,T_0, \qquad \Gamma^{(7,1)}_k=T^O_k\ (k=1,\dots,7), \qquad \Gamma^{(7,1)}_A\Gamma^{(7,1)}_B+\Gamma^{(7,1)}_B\Gamma^{(7,1)}_A=2\,\eta^{(7,1)}_{AB}\,1_{16},$$

$$\Gamma^{(7,1)}_A = S\,\big(i\,\Gamma^{(1,7)}_A\big)\,S^{-1}, \qquad S=\tfrac{1}{\sqrt2}\big(1_{16}-i\,\Gamma^{(1,7)}_0\big), \qquad SS^\dagger=1_{16}.$$

This is literally the formula $T_A=\left(\begin{smallmatrix}0 & \bar\tau_A\\ \tau_A & 0\end{smallmatrix}\right)$ of 4+4.nb, applied to the complexified 8×8 matrices; only the time component is complex.

7.4 Theorem: with $\eta^{(7,1)}$ there is no real 16×16 realization

Theorem. There are no real 16×16 matrices $R_0,\dots,R_7$ with $R_AR_B+R_BR_A=2\eta^{(7,1)}_{AB}1_{16}$. Reordering, renaming or changing the signs of the indices cannot help, because none of these changes the signature $(7,1)$.

Proof. $Cl_8(\mathbb{C})\cong\mathbb{C}(16)$ has exactly one irreducible module, $\mathbb{C}^{16}$. If real R_A existed, R_A and $\Gamma^{(7,1)}_A$ would be two 16-dimensional modules, so $R_A=T\,\Gamma^{(7,1)}_A\,T^{-1}$ for some invertible complex T. Since R_A is real, complex conjugation gives

$$M\,\overline{\Gamma^{(7,1)}_A}=\Gamma^{(7,1)}_A\,M \quad (A=0,\dots,7), \qquad M=T^{-1}\,\overline{T}, \qquad M\,\overline{M}=1_{16}.$$

By Schur's lemma all such M form a one-dimensional space. Solving the linear equations (7.4 of the notebook) gives exactly that space, spanned by the volume element

$$M_0=\Gamma^{(1,7)}_8=\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1\cdots\Gamma^{(1,7)}_7=\begin{pmatrix}0 & -1_8\\ 1_8 & 0\end{pmatrix}, \qquad M_0\,\overline{M_0}=M_0^2=-1_{16}.$$

Hence $M=cM_0$ and $M\overline{M}=-|c|^2\,1_{16}\neq1_{16}$, a contradiction. □ The sign of $M\overline{M}$ is a Frobenius–Schur-type indicator: $+1$ means the complex module has a real form (the algebra is $\mathbb{R}(16)$), -1 means it is quaternionic (the algebra is $\mathbb{H}(8)$, whose smallest real module is $\mathbb{H}^8=\mathbb{R}^{32}$). For $\Gamma^{(1,7)}$ the indicator is $+1$ (take $M=1_{16}$).

7.5 A real realization of $Cl(7,1)$ needs 32×32 matrices

$$\widehat{\Gamma}_A^{(7,1)} = \varepsilon \otimes \Gamma_A^{(1,7)}, \qquad \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \widehat{\Gamma}_A^{(7,1)}\widehat{\Gamma}_B^{(7,1)} + \widehat{\Gamma}_B^{(7,1)}\widehat{\Gamma}_A^{(7,1)} = 2\,\eta_{AB}^{(7,1)}\,1_{32},$$

$$Q_i = \varepsilon \otimes 1_{16}, \qquad Q_j = \sigma_1 \otimes \Gamma_8^{(1,7)}, \qquad Q_k = \sigma_3 \otimes \Gamma_8^{(1,7)},$$

$$Q_i^2 = Q_j^2 = Q_k^2 = -1_{32}, \qquad Q_iQ_j = Q_k = -Q_jQ_i, \qquad Q\,\widehat{\Gamma}_A^{(7,1)} = \widehat{\Gamma}_A^{(7,1)}\,Q \quad (Q = Q_i, Q_j, Q_k).$$

The matrices commuting with all $\widehat{\Gamma}_A^{(7,1)}$ form exactly the 4-dimensional span of $1_{32}, Q_i, Q_j, Q_k$ (computed in the notebook): the quaternions. This is $Cl(7,1) \cong \mathbb{H}(8)$ acting on $\mathbb{R}^{32} = \mathbb{H}^8$.

7.6 Volume elements and chirality

$$T_8^2 = +1_{16} \;\Rightarrow\; P_{L,R} = \tfrac{1}{2}(1_{16} \mp T_8) \text{ real (4+4.nb)}; \qquad (T_8^O)^2 = +1_{16},\; T_8^O = T_8,$$

$$\big(\Gamma_8^{(1,7)}\big)^2 = -1_{16}, \qquad \Gamma_8^{(1,7)}\Gamma_A^{(1,7)} = -\Gamma_A^{(1,7)}\Gamma_8^{(1,7)}, \qquad \Gamma_8^{(1,7)}\Sigma_{AB} = \Sigma_{AB}\Gamma_8^{(1,7)},$$

$$\chi^{(7,1)} = -i\,\Gamma_8^{(7,1)} = T_8^O = T_8, \qquad \tfrac{1}{2}\big(1_{16} \mp \chi^{(7,1)}\big) = P_{L,R},$$

$$\chi^{(1,7)} = i\,\Gamma_8^{(1,7)}, \qquad \tfrac{1}{2}\big(1_{16} \mp \chi^{(1,7)}\big) \text{ complex projectors of rank 8 each.}$$

With real matrices in 7+1 dimensions there is no real chirality projector: $\Gamma_8^{(1,7)}$ is a complex structure commuting with Spin(1,7), so the real 16-component spinor is one complex 8-component Weyl spinor. In the basis of $\Gamma^{(7,1)}$ the chiral projectors are literally P_L and P_R of 4+4.nb.

7.7 What the octonions do: $\Gamma(x)^2 = \mathrm{Re}(xx)$ and alternativity

$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = x^{\mathsf{T}}\eta^{(1,7)}x, \qquad \Gamma^{(1,7)}(x)^2 = \mathrm{Re}(x\,x)\,1_{16},$$

$$T^O(x)^2 = N(x)\,1_{16} = x^{\mathsf{T}}g^O x\,1_{16}, \qquad T(x)^2 = x^{\mathsf{T}}\eta^{(4,4)}x\,1_{16} \text{ (4+4.nb)}.$$

Rotating both 8-blocks by E^O gives $\Gamma^{(1,7)}$ in the octonion basis,

$$\Gamma_0^e = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma_i^e = \begin{pmatrix} L_i^O & 0 \\ 0 & -L_i^O \end{pmatrix}, \qquad \Gamma^e(x)\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \begin{pmatrix} x_0\psi_2 + \vec{x}\,\psi_1 \\ x_0\psi_1 - \vec{x}\,\psi_2 \end{pmatrix} \quad (\psi_1, \psi_2 \in \mathbb{O}),$$

$$\Gamma^e(x)^2\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \begin{pmatrix} x_0^2\psi_1 + \vec{x}(\vec{x}\psi_1) \\ x_0^2\psi_2 + \vec{x}(\vec{x}\psi_2) \end{pmatrix} = (x_0^2 - |\vec{x}|^2)\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix},$$

where $\vec{x} = \sum_i x_i e_i$ and the last step is alternativity, $\vec{x}(\vec{x}\psi) = (\vec{x}\vec{x})\psi = -|\vec{x}|^2\psi$. The Clifford relation of 7+1 spacetime is the alternativity of the octonions.

8. Examples that test everything

8.1 The Lorentz algebra so(1,7)

$$\Sigma_{AB} = \tfrac{1}{4}\big(\Gamma_A^{(1,7)}\Gamma_B^{(1,7)} - \Gamma_B^{(1,7)}\Gamma_A^{(1,7)}\big) \quad (A < B:\ 28 \text{ real, linearly independent matrices}),$$

$$\big[\Sigma_{AB}, \Gamma_K^{(1,7)}\big] = \eta_{BK}^{(1,7)}\,\Gamma_A^{(1,7)} - \eta_{AK}^{(1,7)}\,\Gamma_B^{(1,7)},$$

$$\big[\Sigma_{AB}, \Sigma_{KL}\big] = \eta_{BK}^{(1,7)}\Sigma_{AL} - \eta_{AK}^{(1,7)}\Sigma_{BL} - \eta_{BL}^{(1,7)}\Sigma_{AK} + \eta_{AL}^{(1,7)}\Sigma_{BK}.$$

8.2 A boost along x_1 and a rotation in the x_4–x_5 plane

Because $(\Gamma_0^{(1,7)}\Gamma_1^{(1,7)})^2 = +1_{16}$ and $(\Gamma_4^{(1,7)}\Gamma_5^{(1,7)})^2 = -1_{16}$,

$$S(\beta) = e^{\beta\Sigma_{01}} = \cosh\tfrac{\beta}{2}\,1_{16} + \sinh\tfrac{\beta}{2}\,\Gamma_0^{(1,7)}\Gamma_1^{(1,7)}, \qquad S(\beta)^{-1} = \cosh\tfrac{\beta}{2}\,1_{16} - \sinh\tfrac{\beta}{2}\,\Gamma_0^{(1,7)}\Gamma_1^{(1,7)},$$
$$R(\theta) = e^{\theta\Sigma_{45}} = \cos\tfrac{\theta}{2}\,1_{16} + \sin\tfrac{\theta}{2}\,\Gamma_4^{(1,7)}\Gamma_5^{(1,7)}, \qquad R(\theta)^{-1} = \cos\tfrac{\theta}{2}\,1_{16} - \sin\tfrac{\theta}{2}\,\Gamma_4^{(1,7)}\Gamma_5^{(1,7)},$$

both real (they solve $dS/d\beta = \Sigma_{01}S$, $S(0) = 1_{16}$ and $dR/d\theta = \Sigma_{45}R$, $R(0) = 1_{16}$). They act on the generators as Lorentz transformations, with Λ read off from traces:

$$S\,\Gamma_K^{(1,7)}\,S^{-1} = \sum_{B=0}^{7}\Lambda_{BK}\,\Gamma_B^{(1,7)}, \qquad \Lambda_{BK} = \tfrac{1}{16}\,\mathrm{tr}\big(\eta_{BB}^{(1,7)}\,\Gamma_B^{(1,7)}\,S\,\Gamma_K^{(1,7)}\,S^{-1}\big), \qquad \Lambda^{\mathsf T}\eta^{(1,7)}\Lambda = \eta^{(1,7)},$$

$$\Lambda_{\text{boost}} = \begin{pmatrix} \cosh\beta & -\sinh\beta & 0 \\ -\sinh\beta & \cosh\beta & 0 \\ 0 & 0 & 1_6 \end{pmatrix}, \qquad \Lambda_{\text{rot}} = \begin{pmatrix} 1_4 & 0 & 0 \\ 0 & \begin{smallmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{smallmatrix} & 0 \\ 0 & 0 & 1_2 \end{pmatrix}.$$

8.3 The 7+1 Dirac operator squares to the wave operator

$$\Gamma^A = \eta_{(1,7)}^{AA}\,\Gamma_A^{(1,7)}, \qquad \big(\Gamma^A\partial_A\big)^2\Psi = \eta_{(1,7)}^{AB}\,\partial_A\partial_B\Psi = \big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2\big)\Psi$$

for 16 completely arbitrary functions $\Psi = (\Psi_1, \ldots, \Psi_{16})$ of $x_0, \ldots, x_7$ (the notebook proves it symbolically; here $\eta_{(1,7)}^{AB}$ is the inverse metric, equal to $\eta_{AB}^{(1,7)}$).

8.4 Spinors on the mass shell: real amplitudes

Take $p = (3, 2, 2, 0, 0, 0, 0, 0)$ (contravariant components p^A) and $m = 1$: $p^{\mathsf T}\eta^{(1,7)}p = 9 - 4 - 4 = 1 = m^2$. With $\not{p} = \sum_A p^A\Gamma_A^{(1,7)}$ we have $\not{p}^2 = m^2\,1_{16}$, so $\not{p}$ has eigenvalues $\pm m$, eight times each. Because $\not{p}$ is a real matrix, $(\not{p} - m)w = 0$ has an 8-dimensional space of REAL solutions w (spinor amplitudes). The basis printed by Mathematica (`NullSpace`, one solution per row):

$$W = \begin{pmatrix} 2 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ -2 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & -2 & -2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 2 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -2 & -2 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad (\not{p} - m)\,W^{\mathsf T} = 0.$$

Each row w gives the plane wave $\Psi = w\,e^{-i\,p\cdot x}$, $p\cdot x = p^{\mathsf T}\eta^{(1,7)}x$, solving $(i\Gamma^A\partial_A - m)\Psi = 0$. The plane wave itself is complex: real amplitudes do not make a real field (section 8.5).

8.5 Real spinor fields: massless with 16 components, massive with 32

The operator $i\Gamma^A\partial_A$ is imaginary, so a real Ψ cannot solve $(i\Gamma^A\partial_A - m)\Psi = 0$. A real first-order equation $(G^A\partial_A - m)\Psi = 0$ with real matrices G_A ($G^A = \eta_{(7,1)}^{AA}G_A$) implies the Klein–Gordon equation, through

$$\big(G^A\partial_A + m\big)\big(G^A\partial_A - m\big) = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big),$$

only if $G_AG_B + G_BG_A = 2\,\eta_{AB}^{(7,1)}\,1$ (time squaring to -1), and that relation has no real 16×16 solution (section 7.4). The remaining real option, $(\Gamma^A\partial_A - mJ)\Psi = 0$, squares to the Klein–Gordon operator $\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2$ only if the real matrix J anticommutes with every $\Gamma_A^{(1,7)}$ and $J^2 = +1_{16}$. But (computed in the notebook)

$$\big\{J \in \mathbb{R}(16):\ J\,\Gamma_A^{(1,7)} = -\Gamma_A^{(1,7)}\,J\ \ (A = 0, \ldots, 7)\big\} = \mathbb{R}\,\Gamma_8^{(1,7)}, \qquad \big(c\,\Gamma_8^{(1,7)}\big)^2 = -c^2\,1_{16},$$

and with $J = c\,\Gamma_8^{(1,7)}$ the square is $\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 - c^2m^2$, a tachyon. So with 16 real components only massless real fields exist. For the null momentum $k = (1,1,0,0,0,0,0,0)$, $k^{\mathsf T}\eta^{(1,7)}k = 0$ and $\not{k} = \sum_A k^A\Gamma_A^{(1,7)}$ satisfies $\not{k}^{\,2} = 0$; the real equation $\not{k}\,w = 0$ has 8 real solutions, and for each of them

$$\Psi = w\cos(k\cdot x), \qquad \Gamma^A\partial_A\Psi = -\sin(k\cdot x)\;\; \not{k}\,w = 0 \qquad \text{(a real solution)}.$$

A real massive field needs the real 32×32 matrices $\widehat\Gamma_A^{(7,1)} = \varepsilon\otimes\Gamma_A^{(1,7)}$ of section 7.5, which do satisfy the $\eta^{(7,1)}$ relation. With $\widehat\Gamma^A = \eta^{AA}_{(7,1)}\widehat\Gamma_A^{(7,1)}$ and $P = \sum_A p^A\widehat\Gamma_A^{(7,1)}$ (p as in section 8.4):

$$\big(\widehat\Gamma^A\partial_A\big)^2 = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2\big)\,1_{32}, \qquad P^2 = \big(p^{\mathsf T}\eta^{(7,1)}p\big)\,1_{32} = -m^2\,1_{32},$$
$$\Psi = w\cos(p\cdot x) + \tfrac{1}{m}\,P\,w\,\sin(p\cdot x) \qquad (w\in\mathbb{R}^{32}\text{ arbitrary}),$$
$$\big(\widehat\Gamma^A\partial_A - m\big)\Psi = 0, \qquad \big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big)\Psi = 0.$$

Derivation: with $\phi = p\cdot x$ one has $\partial_A\phi = (\eta^{(1,7)}p)_A$ and $\eta^{AA}_{(7,1)}\eta^{(1,7)}_{AA} = -1$, so $\widehat\Gamma^A\partial_A\Psi = -P\big(-w\sin\phi + \frac{1}{m}Pw\cos\phi\big) = Pw\sin\phi + m\,w\cos\phi = m\Psi$. The 32 real components are the quaternionic module $\mathbb{H}^8 = \mathbb{R}^{32}$ of $Cl(7,1)\cong\mathbb{H}(8)$.

8.6 All nine signatures $p+q=8$

Complex generators of every signature: the first p of the T_A^O and i times the remaining q. The indicator of section 7.4 then decides $\mathbb{R}(16)$ or $\mathbb{H}(8)$ (table of section 4.3). Real 16×16 generators exist exactly in the five $\mathbb{R}(16)$ cases, and all five are built here from the 4+4 and octonion matrices:

$$Cl(8,0):\; T_A^O, \qquad Cl(4,4):\; T_A\;\text{(4+4.nb)}, \qquad Cl(1,7):\; \Gamma_A^{(1,7)},$$
$$Cl(5,3):\; \Gamma_0^{(5,3)} = T_0,\;\; \Gamma_i^{(5,3)} = T_0T_i\;(i=1,\dots,7), \qquad \eta^{(5,3)} = \mathrm{diag}(1,-1,-1,-1,1,1,1,1),$$
$$\Gamma_A^{(5,3)}\Gamma_B^{(5,3)} + \Gamma_B^{(5,3)}\Gamma_A^{(5,3)} = 2\,\eta_{AB}^{(5,3)}1_{16},$$
$$Cl(0,8):\; \Gamma_A^{(0,8)} = T_8^O\,T_A^O, \qquad \Gamma_A^{(0,8)}\Gamma_B^{(0,8)} + \Gamma_B^{(0,8)}\Gamma_A^{(0,8)} = -2\,\delta_{AB}1_{16}.$$

$\Gamma^{(5,3)}$ is the split-octonion analogue of $\Gamma^{(1,7)}$: its quadratic form $x^{\mathsf T}\eta^{(5,3)}x$ is $\mathrm{Re}(xx)$ of the split octonions, just as $x^{\mathsf T}\eta^{(1,7)}x$ is $\mathrm{Re}(xx)$ of the octonions.

9. Verification

In Mathematica. 7+1.nb was evaluated in 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026) with `NotebookEvaluate` and `InsertResults -> True`, in the private context `Notebook$$41$124309`. It contains 145 input cells and 310 checks of the form `check["name", test]`; all 310 returned True, the summary cells printed `failedChecks = {}` and `allChecksPassed = True`, there were 0 message cells, and the evaluation took 225.3 s.

notebook section	checks	True
2. Recap: the 4+4 objects, reproduced exactly from 4+4.nb	24	24
3. The octonions: a real realization mirroring the split octonions	47	47
4. Clifford algebras from the octonions: 16×16 matrices	32	32
5. Examples that test everything	25	25
7. The Dirac equation of the quantized 16-component field dirac16complex	35	35
8. A 16-component Kohn–Sham theory of the self-coupled field dirac16complex	23	23
9. The chiral Kohn–Sham theory (8 + 8 components)	16	16
11. Gravity on the 7+1 spacetime: metric, curvature and the Einstein tensor	7	7
12. The Lovelock tensors of 7+1 spacetime	25	25
13. Generalized frame matrices, the canonical spin connection, and the covariant Dirac equation of dirac16complex	15	15
14. The three epochs: primordial (near x0 = 0), modern, far future	14	14
15. The generalized self-coupling of dirac16complex: all nine ranks	16	16

notebook section	checks	True
16. The Kohn–Sham theory of dirac16complex in gravitational fields	8	8
17. The Standard Model on the world brane of 7+1 spacetime	10	10
18. dirac16complex at the world brane: the 16 components as four brane Dirac spinors, and the portals	7	7
19. Neutrino flavour and flavour conversion	6	6

Independently in Python (exact arithmetic with sympy, in `tools/`): `ref44.py` rebuilds every 4+4.nb object and reproduces the split table of 4+4.nb; `ref71.py` and `extra71.py` rebuild every 7+1 object and re-prove the relations; `verify_7p1_nb.py` parses the evaluated 7+1.nb and confirms that the matrices τ_A, τ_A^O, $\Gamma_A^{(1,7)}$, the two multiplication tables, the matrices Λ, the eight real amplitudes W and the classification table printed by Mathematica coincide with the Python values. For sections 7–9, `ref_dirac16.py` repeats every construction (sympy, mpmath with 30 digits, numpy) with 90 checks, all True, and `verify_7p1_nb.py` compares the printed spinors, the printed closed-form integrals and 23 printed numbers with it (agreement to at least 17 digits) and confirms that sections 0–6 are cell for cell the notebook of the first stage.

For sections 11–19, the independent programs `tools/ref_gravity.py` (99 checks, 99 True; with its own curvature and Lovelock engine `tools/lovelock71.py`), `tools/ref_ks_gravity.py` (29 checks, 29 True) and `tools/ref_sm.py` (32 checks, 32 True) repeat every construction; the Lovelock tensors are computed in the notebook both with xAct (as in the reference notebook Einstein-Lovelock-7+1-Nash.nb) and with an independent pair-set sum, which agree exactly.

10. The notebook 7+1.nb

- **0. How to use this notebook** — evaluation, the `check` bookkeeping, the private context, index conventions.
- **1. Conventions, coordinates and notation** — the coordinate table, the metrics $\eta^{(4,4)}$, $\eta^{(7,1)}$, $\eta^{(1,7)}$, g^O, the dictionary of names, the definitions [1A], [1B] and the four corrections.
- **2. Recap of 4+4.nb** — σ, s_h, t_h, ζ_A, τ_A, $\bar\tau_A$, T_A, P_L, P_R, u, E_{44}, the split table `gridx44` compared with the table printed by 4+4.nb, split product, zero divisors.
- **3. The octonions** — τ^O, vielbein E^O, `grid0`, Fano plane (with a picture), the laws of [1A], the eight-square identity, non-associativity, a numerical example, Cayley–Dickson, comparison with the split octonions.
- **4. Clifford algebras** — T^O ($Cl(8,0)$), $\Gamma^{(1,7)}$ (real 7+1), $\Gamma^{(7,1)}$ (complexified 7+1), the theorem, $\widehat{\Gamma}^{(7,1)}$ (real 32×32) and its quaternionic commutant, chirality, the octonionic meaning.
- **5. Examples** — so(1,7), boost and rotation, Dirac operator, real on-shell spinor amplitudes, real massless (16 components) and massive (32 components) spinor fields, all nine signatures.
- **6. Summary of all checks** of sections 1–5 — `Length[checks]`, `failedChecks`, `allChecksPassed` and a table of the checks.
- **7. The Dirac equation of the quantized 16-component field dirac16complex** — real Dirac matrices, Dirac and Klein–Gordon equations, Hamiltonian, real spinors, spin sums, Grassmann algebra, quantization, charge conjugation, Lorentz covariance, chirality and the two coupled 8-component equations (section 11.1–11.2).
- **8. A 16-component Kohn–Sham theory of the self-coupled field** — Lagrangian, nonlinear field equation, Hohenberg–Kohn, Kohn–Sham equations, exchange (Wick), Fierz, uniform matter (section 11.3, 11.5).
- **9. The chiral Kohn–Sham theory (8 + 8 components)** — chiral densities, chiral Kohn–Sham equations, chiral symmetry, the chiral vacuum with the Dirac sea (section 11.4–11.5).
- **10. Summary of all checks** of sections 1–9.

- **11. Gravity on the 7+1 spacetime** — the curvature engine, the cosmological metric, symmetries and the Bianchi identity (section 12.1).
- **12. The Lovelock tensors** — Lovelock's theorem, the generalized Kronecker delta, xAct (as in Einstein-Lovelock-7+1-Nash.nb), the pair-set algorithm, closed forms, an inhomogeneous metric, a gravitational wave, exponential vacuum solutions (section 12.1).
- **13. Generalized frame matrices, the canonical spin connection and the covariant Dirac equation** — total covariant derivative, spin curvature, Lichnerowicz, Cartan's structure equation, a static radial metric, gauge coupling (section 12.2).
- **14. The three epochs** — Kasner solutions near $x_0 = 0$, pair creation, a static auxiliary space, de Sitter and thermal pair creation (section 12.3).
- **15. The generalized self-coupling** — nine ranks, the Fierz matrix, the covariant nonlinear field equation, chirality (section 12.4).
- **16. Kohn–Sham theory in gravitational fields** — uniform matter in the cosmological spacetime, the epochs, self-consistent Einstein–Lovelock evolution (section 12.4).
- **17. The Standard Model on the world brane** — fields, anomalies, Lagrangian, electroweak breaking, masses and mixing, cosmology (section 12.5).
- **18. dirac16complex at the brane** — four brane Dirac spinors, Kaluza–Klein masses, portals, coupled field equations (section 12.6).
- **19. Neutrino flavour** — vacuum, matter, the Sun, supernovae, the relic background, effects of dirac16complex (section 12.6).
- **20. Summary of all checks** of sections 1–19.

To re-evaluate: open 7+1.nb in Wolfram 15.0.1, start from a fresh kernel (Evaluation ▷ Quit Kernel ▷ Local, if the kernel has been used: the tensor packages xAct of section 12 keep their definitions in the kernel) and choose Evaluation ▷ Evaluate Notebook (about 225 seconds); or run `Get["D:/Developer/github/7+1/tools/run_eval.wl"]` from any notebook, which evaluates 7+1.nb in place in a fresh kernel (a front-end kernel configuration "Eval71" that it adds for the purpose and removes again) and saves it. The notebook is generated by `tools/make_7p1_nb.py` (`python3 tools/make_7p1_nb.py` writes the unevaluated 7+1.nb).

11. Sections 7–10: the Dirac equation and the Kohn–Sham theories of dirac16complex

Sections 7–10 of 7+1.nb were appended in a second stage (sections 0–6 are unchanged). They contain the Dirac equation of a 16-component quantized fermion field named dirac16complex on the real 7+1 spacetime, with eight REAL 16×16 Dirac matrices; its chiral form as two 8-component equations; and the 16-component and chiral Kohn–Sham theories of that field coupled to itself. This section states the results with the notation of this document (section 3); 74 checks of 7+1.nb verify them.

11.1 The Dirac equation with real Dirac matrices

The field is the column $\Psi = (\Psi_1, \ldots, \Psi_{16})^{\mathsf{T}}$ (`dirac16complex`) of complex anticommuting (Grassmann-odd) operators, with the equal-time anticommutators

$$\Psi_a(x_0,\vec{x})\Psi_b(x_0,\vec{y}) + \Psi_b(x_0,\vec{y})\Psi_a(x_0,\vec{x}) = 0, \qquad \Psi_a(x_0,\vec{x})\Psi_b^\dagger(x_0,\vec{y}) + \Psi_b^\dagger(x_0,\vec{y})\Psi_a(x_0,\vec{x}) = \delta_{ab}\,\delta^7(\vec{x}-\vec{y}),$$

where $\vec{x} = (x_1, \ldots, x_7)$ here denotes the space coordinates. The Dirac matrices are the real generators of section 7.2, $\Gamma_0^{(1,7)} = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i^{(1,7)} = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$, with $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A^{(1,7)}$ and $\bar\Psi = \Psi^\dagger\Gamma^0$:

$$\big(i\,\Gamma^A\partial_A - m\big)\Psi = 0, \qquad \big(i\Gamma^A\partial_A + m\big)\big(i\Gamma^A\partial_A - m\big) = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big)1_{16},$$
$$i\,\partial_0\Psi = H_{\mathrm{D}}\Psi, \qquad H_{\mathrm{D}} = -i\,\Gamma^0\Gamma^i\partial_i + m\,\Gamma^0, \qquad \partial_A\big(\bar\Psi\Gamma^A\Psi\big) = 0.$$

Γ^0 and $\Gamma^0\Gamma^i$ are real symmetric, so $H_{\rm D}$ is Hermitian. The factor i is required: with real matrices, $(\Gamma^A\partial_A)^2 = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2$, and only $i\Gamma^A\partial_A$ gives the Klein–Gordon operator with $+m^2$. For a plane wave $e^{-i(Ex_0-\vec{p}\cdot\vec{x})}$, $H_{\rm D}$ becomes the real matrix $\left(\begin{smallmatrix} 0 & m1_8+p^i\tau_i^O \\ m1_8-p^i\tau_i^O & 0 \end{smallmatrix}\right)$, and since $(m1_8+p^i\tau_i^O)(m1_8-p^i\tau_i^O) = E^2 1_8$, $E = \sqrt{m^2+\vec{p}^{\,2}}$, its positive-energy eigenvectors are REAL: $w_s = \frac{1}{\sqrt{E}}\big(E\,e_s;\ (m1_8 - p^i\tau_i^O)e_s\big)$, $s = 1,\ldots,8$ (e_s the unit columns of length 8), real amplitudes with $(\not{p} - m)w_s = 0$ as in section 8.4. The notebook also verifies the spin sums, the mode expansion with the canonical anticommutators, charge conjugation $\Psi^c = \Gamma_8^{(1,7)}\Psi^*$ and Lorentz covariance.

11.2 Chiral projections: two coupled 8-component equations

There is no real chirality matrix (section 8.5: the real matrices anticommuting with all $\Gamma_A^{(1,7)}$ are the multiples of $\Gamma_8^{(1,7)}$, which squares to -1_{16}). With the complex chirality $\chi^{(1,7)} = i\Gamma_8^{(1,7)}$ (section 7.6) the chiral projections $P_{L,R}^{(1,7)} = \frac{1}{2}\big(1_{16} \mp \chi^{(1,7)}\big)$ give, with $\Psi = (\Psi_\uparrow;\Psi_\downarrow)$ (upper and lower 8 components),

$$\Psi_L = \frac{\Psi_\uparrow + i\,\Psi_\downarrow}{\sqrt{2}}, \qquad \Psi_R = \frac{i\,\Psi_\uparrow + \Psi_\downarrow}{\sqrt{2}},$$

$$P_L^{(1,7)}\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} 1_8 \\ -i1_8 \end{pmatrix}\Psi_L, \qquad P_R^{(1,7)}\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} -i1_8 \\ 1_8 \end{pmatrix}\Psi_R,$$

$$\big(i\,\partial_0 + \tau_i^O\,\partial_i\big)\Psi_L = m\,\Psi_R, \qquad \big(i\,\partial_0 - \tau_i^O\,\partial_i\big)\Psi_R = m\,\Psi_L.$$

The two 8-component equations contain only the real matrices τ_i^O; the mass couples them, and for $m = 0$ they decouple.

11.3 The self-coupled field and its 16-component Kohn–Sham theory

With $\chi = \chi^{(1,7)}$, four real couplings g_S, g_P, g_V, g_A and four real external sources $\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$:

$$\begin{aligned}\mathcal{L} = {}& \bar\Psi\big(i\Gamma^A\partial_A - m\big)\Psi - \bar\Psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\Psi \\ & - \frac{g_S}{2}(\bar\Psi\Psi)^2 - \frac{g_P}{2}(\bar\Psi i\chi\Psi)^2 - \frac{g_V}{2}(\bar\Psi\Gamma^A\Psi)(\bar\Psi\Gamma_A\Psi) - \frac{g_A}{2}(\bar\Psi\Gamma^A\chi\Psi)(\bar\Psi\Gamma_A\chi\Psi),\end{aligned}$$

where $\Gamma_A = \Gamma_A^{(1,7)}$. The densities $\rho_S = \bar\Psi\Psi$, $\rho_P = \bar\Psi i\chi\Psi$, $j^A = \bar\Psi\Gamma^A\Psi$, $j_5^A = \bar\Psi\Gamma^A\chi\Psi$ (18 real fields) determine the ground state (Hohenberg–Kohn), and the Kohn–Sham orbitals Φ_k obey

$$\begin{aligned} H_{\rm KS}\Phi_k &= \lambda_k\Phi_k, \\ H_{\rm KS} &= \Gamma^0\Gamma^i\big(-i\partial_i + \mathcal{V}_i\big) + \Gamma^0\big(m + \mathcal{S}\big) + \mathcal{V}_0 + \Gamma^0 i\chi\,\mathcal{P} + \chi\,\mathcal{A}_0 + \Gamma^0\Gamma^i\chi\,\mathcal{A}_i, \\ \mathcal{S} &= \mathsf{s} + g_S\rho_S + \frac{\delta E_{xc}}{\delta\rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\rho_P + \frac{\delta E_{xc}}{\delta\rho_P}, \\ \mathcal{V}_A &= \mathsf{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathsf{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A}, \\ \rho_S &= \sum_k \bar\Phi_k\Phi_k, \quad \rho_P = \sum_k \bar\Phi_k i\chi\Phi_k, \quad j^A = \sum_k \bar\Phi_k\Gamma^A\Phi_k, \quad j_5^A = \sum_k \bar\Phi_k\Gamma^A\chi\Phi_k, \end{aligned}$$

with $j_A = \eta_{AA}^{(1,7)} j^A$, the sums running over the occupied positive-energy orbitals (no Dirac sea), and $E_{xc} = F - T_s - E_H$, where F is the universal functional, $T_s = \sum_k \int \Phi_k^\dagger H_{\rm D}\Phi_k$ and $E_H = \int\big[\frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2}j^A j_A + \frac{g_A}{2}j_5^A j_{5A}\big]$. The exact exchange energy of the contact couplings follows from Wick's theorem: with the 16×16 matrix $\mathcal{R} = \sum_k \Phi_k\bar\Phi_k$,

$$E_x = -\frac{g_S}{2}\mathrm{tr}(\mathcal{R}\mathcal{R}) - \frac{g_P}{2}\mathrm{tr}(i\chi\mathcal{R}\,i\chi\mathcal{R}) - \frac{g_V}{2}\mathrm{tr}(\Gamma^A\mathcal{R}\Gamma_A\mathcal{R}) - \frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi\mathcal{R}\Gamma_A\chi\mathcal{R}).$$

11.4 The chiral Kohn–Sham theory

In chiral components Φ_{kL}, Φ_{kR} (as for Ψ_L, Ψ_R), with $\kappa=\sum_k \Phi_{kL}^\dagger\Phi_{kR}$ ($\rho_S=\kappa+\kappa^*$, $\rho_P=i(\kappa-\kappa^*)$), the complex Kohn–Sham mass $\mathcal{M}=m+\mathcal{S}+i\mathcal{P}$ and $\mathcal{V}_A^{L,R}=\mathcal{V}_A\mp\mathcal{A}_A$, the Kohn–Sham equation becomes

$$\big[\mathcal{V}_0^L-\tau_i^O\partial_i-i\tau_i^O\mathcal{V}_i^L\big]\Phi_L+\mathcal{M}\,\Phi_R=\lambda\,\Phi_L,$$
$$\mathcal{M}^*\,\Phi_L+\big[\mathcal{V}_0^R+\tau_i^O\partial_i+i\tau_i^O\mathcal{V}_i^R\big]\Phi_R=\lambda\,\Phi_R.$$

The scalar–pseudoscalar Hartree energy $\frac{g_S-g_P}{2}(\kappa^2+\kappa^{*2})+(g_S+g_P)\kappa\kappa^*$ is invariant under the real chiral rotations $e^{i\vartheta\chi}=\cos\vartheta\,1_{16}-\sin\vartheta\,\Gamma_8^{(1,7)}$ ($\kappa\to e^{2i\vartheta}\kappa$) exactly when $g_P=g_S$.

11.5 Examples and verification

Uniform matter (units $m=1$, no sea, $g_S=-10^5$, $g_V=10^4$, $g_P=g_A=0$): the Hartree gap equation $m^*=m+g_S\rho_S(m^*)$ and the energy density give saturation at the Fermi momentum $k_F=0.756636150702$, where the energy per particle is $E/A-m=-0.0283214279647$ (bound) and $m^*=0.348097340146$; there E/A equals the Fermi energy $E_F=g_V\rho+\sqrt{k_F^2+m^{*2}}$ (Hugenholtz–Van Hove), with $\rho=k_F^7/(105\pi^4)$.

The chiral vacuum with the Dirac sea (all negative-energy orbitals filled, momentum cutoff $|\vec p|<\Lambda_{\rm c}=1$, $g_S=g_P=g$): for $m=0$ a non-zero mass $|\mathcal{M}|$ appears exactly when $g<g_c=-90\pi^4/\Lambda_{\rm c}^6$; for $g=\frac{6}{5}g_c$, $|\mathcal{M}|=0.553693851829$ — spontaneous breaking of the chiral symmetry.

Checks. Sections 7–9 of 7+1.nb contain 74 checks, all True (section 9); the independent program `tools/ref_dirac16.py` repeats the constructions with 90 checks, all True, and reproduces every printed number to at least 17 digits.

12. Sections 11–19: gravity, the self-coupled dirac16complex in gravitational fields, and the Standard Model

Sections 11–19 of 7+1.nb (and the summary section 20) were appended in a third stage; sections 0–10 are unchanged. They rebuild the ideas of the repository Dirac_claude on the 7+1 spacetime of this project — ONE time coordinate x_0 and SEVEN space coordinates $x_1,\dots,x_7$ (x_1,x_2,x_3 ordinary space, $x_4,\dots,x_7$ auxiliary space); the 4+4 metric of Dirac_claude is not used. Here $\Gamma_A=\Gamma_A^{(1,7)}$, $\Gamma^A=\eta^{AA}_{(1,7)}\Gamma_A$, $\eta=\eta^{(1,7)}=\mathrm{diag}(+1,-1,\dots,-1)$, $\chi=i\Gamma_0\Gamma_1\cdots\Gamma_7$, Ψ is the field dirac16complex and $\bar\Psi=\Psi^\dagger\Gamma_0$; a prime is d/dx_0. The sections contain 97, 24 and 23 checks (gravity, Kohn–Sham, Standard Model), of which 97, 24 and 23 are True.

12.1 Curvature and the Lovelock tensors (sections 11–12)

Christoffel symbols are written $\{{}^{\lambda}_{\mu\nu}\}$ (Γ always denotes a Dirac matrix); the curvature follows Landau and Lifshitz:

$$\{{}^{\lambda}_{\mu\nu}\}=\tfrac12 g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu}+\partial_\nu g_{\rho\mu}-\partial_\rho g_{\mu\nu}\big),\qquad R^\rho{}_{\sigma\mu\nu}=\partial_\mu\{{}^{\rho}_{\sigma\nu}\}-\partial_\nu\{{}^{\rho}_{\sigma\mu}\}+\{{}^{\rho}_{\mu\lambda}\}\{{}^{\lambda}_{\sigma\nu}\}-\{{}^{\rho}_{\nu\lambda}\}\{{}^{\lambda}_{\sigma\mu}\},$$
$$R_{\sigma\nu}=R^\rho{}_{\sigma\rho\nu},\qquad R=g^{\sigma\nu}R_{\sigma\nu},\qquad G^\mu{}_\nu=R^\mu{}_\nu-\tfrac12\delta^\mu_\nu R,\qquad R^{\gamma\delta}{}_{\alpha\beta}=g^{\delta\sigma}R^\gamma{}_{\sigma\alpha\beta}.$$

In n dimensions the most general symmetric, divergence-free tensor built from the metric and its first two derivatives is a sum of $m-1$ Lovelock tensors plus a cosmological term, $m=n/2$ for even n (D. Lovelock and H. Rund): for $n=8$ there are exactly three, normalized as in the reference notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, GPL),

$$E^{(k)\mu}{}_\nu=-\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\,R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},\qquad L_k=\frac{1}{2^k}\,\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k}\,R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$
$$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n}=\det\big[\delta^{\mu_i}_{\nu_j}\big],\qquad E^{(1)}=G,\qquad E^{(k)\mu}{}_\mu=-\frac{8-2k}{2}L_k,\qquad E^{(4)}\equiv0.$$

The field equations, in the form of the reference notebook (pure numbers w_k, Λ and an inverse length H) and with matter ($\kappa=8\pi G_8$):

$$0=-\Lambda\,\delta^\mu_\nu+\sum_{k=1}^{3}H^{-2k}w_k\,E^{(k)\mu}{}_\nu,\qquad\sum_{k=0}^{3}\alpha_k E^{(k)\mu}{}_\nu=\kappa T^\mu{}_\nu\ \ (\alpha_0=2\bar\Lambda,\ \alpha_1=1),$$

$$\alpha_2 = \frac{w_2}{w_1 H^2}, \quad \alpha_3 = \frac{w_3}{w_1 H^4}, \quad \bar{\Lambda} = \frac{\Lambda H^2}{w_1}.$$

The notebook computes the Lovelock tensors twice: with the tensor packages xAct (xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2) exactly as the reference notebook does — abstract indices, the generalized delta `kδ[lower, upper] = Det[Outer[delta, lower, upper]]`, and the components of a given metric from `CD = LC[g]` — and by an independent sum over sets of index pairs; the two agree exactly for the cosmological metric and a gravitational wave. The reference notebook's section "FAILED Model metric g71; DO NOT USE!" is not used. For the cosmological metric $ds^2 = dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2)$ with $H_a = a'/a$, $H_b = b'/b$:

$$E^{(1)0}{}_0 = 3H_a^2 + 12H_aH_b + 6H_b^2, \qquad E^{(2)0}{}_0 = -12H_b\big(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3\big),$$
$$E^{(3)0}{}_0 = 360\, H_a^2H_b^3\big(4H_a + 3H_b\big).$$

Exponential vacuum solutions ($a = e^{Hx_0}$, $b = e^{q_H Hx_0}$) exist for a one-parameter family of couplings; for $q_H = -1$ (eternal inflation of ordinary space with eternal deflation of the auxiliary space) $w_2 = \frac{\Lambda}{4} - \frac{w_1}{2}$, $w_3 = -\frac{\Lambda}{36} + \frac{w_1}{24}$.

12.2 Generalized frame matrices, the canonical spin connection, and the covariant Dirac equation (section 13)

With a frame field $g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu$ and its inverse $e_A{}^\mu$:

$$\Gamma_\mu(X) = e^A{}_\mu \Gamma_A, \qquad \Gamma^\mu(X) = e_A{}^\mu \Gamma^A, \qquad \Gamma_\mu\Gamma_\nu + \Gamma_\nu\Gamma_\mu = 2g_{\mu\nu}1_{16},$$
$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\} e_B{}^\lambda\big), \qquad \Omega_\mu = \tfrac{1}{4}\eta_{AC}\,\omega_\mu{}^C{}_B\,\Gamma^A\Gamma^B, \qquad D_\mu = \partial_\mu + \Omega_\mu,$$
$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{{}^\lambda_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0, \qquad [D_\mu, D_\nu] = \tfrac{1}{4}R_{AB\mu\nu}\Gamma^A\Gamma^B, \qquad (\Gamma^\mu D_\mu)^2 = g^{\mu\nu}\nabla_\mu D_\nu - \tfrac{R}{4}.$$

The covariant Dirac equation $\big(i\Gamma^\mu(X)D_\mu - m\big)\Psi = 0$ follows from $\mathcal{L} = \sqrt{|g|}\,\bar\Psi(i\Gamma^\mu D_\mu - m)\Psi$, and $\bar\Psi\Gamma^\mu\Psi$ is conserved. For the cosmological metric the spin connection is not zero:

$$\Omega_0 = 0, \quad \Omega_i = \tfrac{a'}{2}\Gamma_i\Gamma_0 \ (i = 1,2,3), \quad \Omega_k = \tfrac{b'}{2}\Gamma_k\Gamma_0 \ (k = 4,\ldots,7), \qquad \Gamma^\mu\Omega_\mu = \big(\tfrac{3}{2}H_a + 2H_b\big)\Gamma_0.$$

The notebook also derives the connection from Cartan's structure equation $de^A + \omega^A{}_B \wedge e^B = 0$ for a static metric with a radial coordinate, and couples an abelian gauge field, $D_\mu = \partial_\mu + \Omega_\mu - iqA_\mu$.

12.3 The three epochs (section 14)

Primordial (near $x_0 = 0$): for $a = x_0^p$, $b = x_0^q$ the highest Lovelock order dominates. Einstein: $3p + 4q = 1$, $3p^2 + 4q^2 = 1$, $p = (1 \pm 2\sqrt{2})/7$, $q = (2 \mp 3\sqrt{2})/14$ — on the branch $p = 0.547$, $q = -0.160$ ordinary space expands while the auxiliary space contracts. Gauss–Bonnet: $3p + 4q = 3$, $-203p^3 + 81p^2 + 351p + 27 = 0$ ($p = -1.08350, -0.0786311, 1.56115$). Cubic: $(p, q) = (-15/7, 20/7)$. The anisotropic expansion creates pairs of dirac16complex. **Modern**: with $b = b_0$ the field equations reduce to the Friedmann equations of ordinary space, $3H_a^2 = \kappa\varepsilon + \bar\Lambda$, $2H_a' + 3H_a^2 = -\kappa p_a + \bar\Lambda$, with $G_4 = G_8/V_\perp$. **Far future**: in de Sitter space pairs are created with the Fermi–Dirac occupation at the Gibbons–Hawking temperature $H/2\pi$:

$$|\beta|^2 = \frac{1}{e^{2\pi M/H} + 1}.$$

12.4 The nine-channel self-coupling and Kohn–Sham theory in gravitational fields (sections 15–16)

With $\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}$ for the 256 increasing multi-indices K ($\sigma_r = \frac{r(r-1)}{2} \bmod 2$, $\Theta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}\Theta^K$) the most general contact self-coupling has one coupling per rank,

$$\mathcal{L}_{\text{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r \sum_{|K|=r} (\bar\Psi\Theta_K\Psi)(\bar\Psi\Theta^K\Psi), \qquad \big(i\Gamma^\mu D_\mu - m\big)\Psi = \sum_r g_r \sum_{|K|=r} (\bar\Psi\Theta_K\Psi)\,\Theta^K\Psi,$$

and contains the four couplings of section 11.3 ($g_0 = g_S$, $g_1 = g_V$, $g_7 = -g_A$, $g_8 = -g_P$). The Fierz matrix $F_{rs} = \frac{1}{16}\sum_l (-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l}$ ($F^2 = 1$) turns exchange into direct terms, so Hartree–Fock is Hartree with $G = (1 - F^{\mathsf{T}})g$. Homogeneous Kohn–Sham matter in the cosmological spacetime has the effective mass $M = m + G_0\rho_S$, the energy density ε and the pressures p_a, p_b (integrals over the comoving Fermi ball), and

$$\varepsilon' + 3H_a(\varepsilon + p_a) + 4H_b(\varepsilon + p_b) = 0.$$

Near $x_0 = 0$ the effective mass falls toward zero ($M = 0.003824$ at $x_0 = 10^{-4}$), in the modern epoch the matter is pressureless in ordinary space, and in a de Sitter future $1 - M$ decays like the density; Einstein and Einstein–Lovelock gravity driven by this matter preserve the constraint equation to a relative 2.4×10^{-14}.

12.5 The Standard Model on the world brane (section 17)

The auxiliary coordinates form a torus ($x_k \simeq x_k + L_c$); the Standard Model lives on the brane $W = \{x_4 = \cdots = x_7 = 0\}$ with the induced metric $h_{\mu\nu} = g_{\mu\nu}(x, y = 0)$, and the total action is

$$S = S_{\mathrm{EL}}[g] + S_\Psi[\Psi, g] + \int_W d^4x\sqrt{|h|}\,\big(\mathcal{L}_{\mathrm{SM}} + \mathcal{L}_{\mathrm{portal}}\big) + S_{\mathrm{stab}},$$

$$S_{\mathrm{EL}} = -\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\sum_{k=0}^{3}\alpha_k L_k, \qquad S_\Psi = \int d^8X\sqrt{|g|}\,\big[\bar\Psi(i\Gamma^\mu D_\mu - m)\Psi + \mathcal{L}_{\mathrm{self}}\big],$$

(S_{stab} is a stress that keeps the auxiliary scale factor b constant in the modern epoch), with the complete Standard Model ($SU(3) \times SU(2) \times U(1)$, three generations with right-handed neutrinos, one Higgs doublet, Yukawa couplings) in the gravitational field of h. All gauge anomalies cancel in every generation; with the data of the Particle Data Group the electroweak scale is $v = (\sqrt{2}G_F)^{-1/2} = 246.219640$ GeV and the CKM Jarlskog invariant $J = 3.117 \times 10^{-5}$; the brane Friedmann equation $3H_a^2 = 8\pi G_4\rho_4 + \Lambda_4$ with the Planck 2018 parameters gives the age 13.791 Gyr.

12.6 dirac16complex at the brane, portals, and neutrino flavour (sections 18–19)

With the Dirac representation γ_μ ($\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\eta_{\mu\nu}1_4$), $\gamma_5 = i\gamma_0\gamma_1\gamma_2\gamma_3$ and $\rho_4 = i\gamma_0$, $\rho_5 = -\gamma_1$, $\rho_6 = -\gamma_2$, $\rho_7 = -\gamma_3$, the matrices

$$R_\mu = \gamma_\mu \otimes 1_4\ (\mu = 0, \ldots, 3), \qquad R_k = \gamma_5 \otimes \rho_k\ (k = 4, \ldots, 7), \qquad \Gamma_A U = U R_A, \quad U^\dagger U = 1_{16},$$

show that dirac16complex at the brane is four ordinary Dirac spinors, $\Psi = U\sum_a \psi_a \otimes e_a$, with $\bar\Psi\Gamma^\mu\Psi' = \sum_a \bar\psi_a\gamma^\mu\psi'_a$; a mode with auxiliary momentum $\vec q$ has the Kaluza–Klein mass $\sqrt{m^2 + |\vec q|^2/b_0^2}$. Three portals (Yukawa, vector, Higgs) couple it to the Standard Model at the brane; the Yukawa portal makes the light neutrino mixing non-unitary, $1 - NN^\dagger = \mu\mu^\dagger/m^2$. Neutrino flavour conversion in vacuum,

$$P(\alpha \to \beta) = \delta_{\alpha\beta} - 4\sum_{i>j}\mathrm{Re}\,X_{ij}\sin^2\frac{\Delta m^2_{ij}L}{4E} + 2\sum_{i>j}\mathrm{Im}\,X_{ij}\sin\frac{\Delta m^2_{ij}L}{2E}, \qquad X_{ij} = U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j},$$

in matter (potential $\sqrt{2}G_F n_e$), in the Sun and in supernovae is worked out with NuFit-6.0 data; for DUNE ($L = 1300$ km, $E = 2.5$ GeV) $P(\nu_\mu \to \nu_e) = 0.048837$ in vacuum and 0.072327 in the Earth's crust.

Checks. Sections 11–19 of 7+1.nb contain 144 checks, all True; the independent programs `tools/ref_gravity.py`, `tools/ref_ks_gravity.py` and `tools/ref_sm.py` repeat the constructions with 99, 29 and 32 checks, all True, and reproduce the printed numbers.

13. Files in this repository

path	content
`7+1.nb`	the new notebook, evaluated, with all outputs
`4+4.nb`	the 4+4 notebook (split octonions, T_A), the starting point; unchanged
`NASH_Mathematica_Notebooks/`	earlier notebooks of the project; unchanged
`tools/make_7p1_nb.py`, `tools/nbwrite.py`	generator of 7+1.nb
`tools/run_eval.wl`	evaluates 7+1.nb in place in a fresh kernel of Wolfram 15.0.1 and saves it
`tools/watch_requests.wl`, `tools/git_push.wl`	runs Wolfram 15.0.1 jobs requested in build/request.txt (evaluate, push) without keyboard input; pushes with the local Git
`tools/eval_cells_log.wl`	development check: evaluates a generated notebook cell by cell and logs messages and checks
`tools/ref_dirac16.py`	independent Python checks of sections 7–9 (dirac16complex, Kohn–Sham, chiral Kohn–Sham)
`tools/ref44.py`, `tools/ref71.py`, `tools/extra71.py`, `tools/octalg.py`	independent Python (sympy) implementation and checks
`tools/classify_clifford.py`	numerical classification of the nine signatures via commutants of realified representations
`tools/cayley_dickson_formulas.py`, `tools/cayley_dickson_match.py`	tests of Cayley–Dickson formulas against the tables
`tools/nbparse.py`, `tools/verify_7p1_nb.py`	parser for .nb files; cross-check of the evaluated 7+1.nb
`tools/docgen.py`, `tools/doc_*.py`, `tools/make_docs.py`	generators of the documentation (Markdown, LaTeX, PDF)
`tools/nb_part_gravity.py`, `tools/nb_part_ks_gravity.py`, `tools/nb_part_sm.py`	generator parts of sections 11–19 of 7+1.nb (executed by `make_7p1_nb.py`)
`tools/lovelock71.py`	independent curvature and Lovelock engine (pair-set algorithm, sympy)
`tools/ref_gravity.py`, `tools/ref_ks_gravity.py`, `tools/ref_sm.py`	independent Python checks of sections 11–14, 15–16 and 17–19

14. References

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